

(...), then {...} and then [...].

* For modulus e.g. $|-3|$ we write magnitude only not sign i.e. 3 in this case.

Important Formulas

1. $(a + b)^2 = a^2 + 2ab + b^2$
2. $(a - b)^2 = a^2 - 2ab + b^2$
3. $(a^2 - b^2) = (a + b)(a - b)$
4. $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
5. $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
6. $(a^3 + b^3) = (a + b)(a^2 - ab + b^2)$
7. $(a^3 - b^3) = (a - b)(a^2 + ab + b^2)$
8. $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = \frac{1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2]$
Here, if $(a + b + c) = 0$,
then $a^3 + b^3 + c^3 - 3abc = 0$
 $\Rightarrow a^3 + b^3 + c^3 = 3abc$
9. $(a + b + c)^2 = (a^2 + b^2 + c^2) + 2(ab + bc + ca)$

Variety Questions

Q.1. Simplify $\frac{(25)^4 + 1250 + 1 - (25)^2}{(25)^2 + 25 + 1}$.
SSC CGL 19/09/2024 (3rd Shift)
(a) 602 (b) 601 (c) 603 (d) 604

Q.2. Simplify $\frac{25a^2 - 10ab - 48b^2}{5a + 6b} \times (5a + 8b)$
SSC CGL 13/09/2024 (2nd Shift)
(a) $25a^2 - 64b^2$ (b) $(5a - 8b)^2$
(c) $(5a + 8b)^2$ (d) $25a^2 - 36b^2$

Q.3. The simplified form of the expression $\frac{(x^2 - 9) \times (9x^2 - 1)}{(x - \frac{1}{3}) \times (1 - \frac{x}{3})} \times \frac{9}{(x + \frac{1}{3})(1 + \frac{x}{3})}$ is _____.
SSC CHSL Tier II (10/01/2024)
(a) -729 (b) -81 (c) 729 (d) 81

Q.4. If $A = \frac{58^2 - 25^2}{46^2 - 37^2}$, $B = \frac{26^2 - 15^2}{56^2 - 15^2}$,
then the value of $\frac{1}{B} - \frac{20}{A}$ is :
SSC CGL Tier II (26/10/2023)
(a) 2 (b) 1 (c) -1 (d) 0

Q.5. On comparing the following two numeric expressions, we find that _____.
 $[(\frac{7}{9})^2 \cdot \frac{1}{2}]^{\frac{3}{5}}$ and $[(1\frac{2}{3})^9]^{\frac{3}{5}}$
SSC MTS 14/09/2023 (2nd Shift)
(a) both the expressions are equal

- (b) the first expression is smaller than the second
- (c) the first expression is larger than the second
- (d) the given two expressions cannot be compared

Q.6. On simplifying the following expression, we get _____.

$$\frac{24\sqrt{45} - 16\sqrt{20} + \sqrt{245} - 47\sqrt{5}}{3\frac{1}{5} + 1\frac{2}{5} - 2\frac{1}{6} - 2\frac{1}{5} - 5\frac{2}{5} + 3\frac{1}{6}}$$

SSC MTS 08/09/2023 (2nd Shift)
(a) 0 (b) 3 (c) 1 (d) 2

Q.7. Simplify :

$$\frac{(3.321)^3 + (2.681)^3 + (1.245)^3 - 3 \times 3.321 \times 2.681 \times 1.245}{(3.321)^2 + (2.681)^2 + (1.245)^2 - (3.321 \times 2.681) - (2.681 \times 1.245) - (1.245 \times 3.321)}$$

= ?
SSC CHSL 04/08/2023 (1st Shift)
(a) 6.125 (b) 8.645 (c) 7.247 (d) 10.245

Q.8. Simplify the following expression.
 $0.01 + 0.68 \times 0.68 + 0.22 \times 0.22 + 0.044 + 0.136 + 0.44 \times 0.68$
SSC CHSL 03/08/2023 (1st Shift)
(a) 0.69 (b) 1 (c) 0.8 (d) 0.9

Q.9. Simplify the given expression
 $\frac{0.09 \times 0.09 + 0.04 \times 0.04 + 0.16 \times 0.16 + 2 \times 0.09 \times 0.04 + 2 \times 0.04 \times 0.16 + 2 \times 0.09 \times 0.16}{0.3 \times 0.3 + 0.2 \times 0.2 + 0.4 \times 0.4}$
SSC CHSL 02/08/2023 (4th Shift)
(a) 0.38 (b) 0.24 (c) 0.32 (d) 0.29

Q.10. What is the value of the given Expression ?
 $\frac{4^{a+4} - 5 \times 4^{a+2}}{15 \times 4^a - 2^2 \times 4^a}$
SSC CGL 24/07/2023 (1st shift)
(a) 16 (b) 64 (c) 20 (d) 24

Q.11. What is the value of $(2^2 + 1)(2^4 + 1)(2^8 + 1)(2^{16} + 1) \dots (2^{128} + 1)$?
SSC CHSL 14/03/2023 (2nd Shift)
(a) $\frac{2^{256} - 1}{2}$ (b) $\frac{2^{256} - 1}{3}$
(c) $2^{256} - 1$ (d) $\frac{2^{256} - 1}{4}$

Q.12. Which of the following statement is true ?
I. $\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \dots + \frac{1}{110} < \frac{5}{6}$
II. $\frac{1}{3} + \frac{1}{15} + \frac{1}{35} + \dots + \frac{1}{143} > \frac{7}{13}$
SSC CHSL 13/03/2023 (4th Shift)
(a) Only I (b) Both I and II
(c) Only II (d) Neither I nor II

Q.13. Find the value of the following expression.
 $[\frac{5}{8} - \{\frac{3}{8} - (\frac{5}{8} - \frac{3}{8})\}] \text{ of } 8.8 - 1.2$
 $\frac{4\frac{1}{6} \div 2.5 \times 2 \div \frac{1}{6} \text{ of } 60 + (\frac{3}{4} - \frac{3}{8})}{}$
SSC CPO 11/11/2022 (Morning)
(a) $5\frac{22}{43}$ (b) $3\frac{23}{67}$ (c) $4\frac{44}{85}$ (d) $4\frac{4}{5}$

Q.14. Solve the following.

$$24 \div \frac{3}{8} \text{ of } (8 + 2 \times \overline{7-3}) + [\frac{2}{11} \div \frac{4}{55} - \{\frac{5}{8} + \frac{6}{16}\}]$$

$$32 \div \overline{15-7} + 75 \div (6 + 15 \div 3 + 4)$$

SSC CPO 09/11/2022 (Morning)
(a) $\frac{23}{27}$ (b) $\frac{9}{2}$ (c) $\frac{11}{18}$ (d) $\frac{15}{19}$

Q.15. Solve

$$\frac{(6.9)^3 + (2.7)^3 + (5.8)^3 - 3 \times 6.9 \times 2.7 \times 5.8}{(6.9)^2 + (2.7)^2 + (5.8)^2 - (6.9)(2.7) - (2.7)(5.8) - (5.8)(6.9)}$$

SSC CHSL 07/06/2022 (Afternoon)
(a) 15.1 (b) 15.3 (c) 15.2 (d) 15.4

Q.16. Simplify the following expression.

$$9\frac{1}{5} \div \{(16\frac{1}{5} \div 12\frac{2}{3} - 9\frac{2}{3}) + 17\frac{1}{2} \text{ of } 3\frac{1}{21}\}$$

SSC CHSL 26/05/2022 (Evening)
(a) $\frac{139}{881}$ (b) $\frac{137}{881}$ (c) $\frac{140}{881}$ (d) $\frac{138}{881}$

Q.17. The value of $\frac{1}{4} + \frac{[(20.35)^2 - (8.35)^2] \times 0.0175}{(1.05)^2 + (1.05)(27.65)}$ is:
SSC CGL Tier II (29/01/2022)
(a) $\frac{3}{20}$ (b) $\frac{3}{10}$ (c) $\frac{7}{20}$ (d) $\frac{9}{20}$

Q.18. The simplified value of $\frac{119 - [48 \div 6 - 7 \{5 \times 12 \div 3 - (15 - \overline{3-8})\}]}{24 \div 3 - 9 \text{ of } 3 + (52 - 8 - 4) \div 6}$ is:
SSC MTS 11/10/2021 (Morning)
(a) $-\frac{111}{11}$ (b) $-\frac{127}{27}$ (c) $\frac{127}{43}$ (d) $\frac{111}{11}$

Q.19. If $\frac{1}{4 - \sqrt{8}} + \frac{3 + 2\sqrt{2}}{3 - 2\sqrt{2}} - \frac{3 - 2\sqrt{2}}{3 + 2\sqrt{2}} = a + b\sqrt{2}$, then what is the value of $(3a + 4b)$?
SSC CGL Tier II (15/11/2020)
(a) $99\frac{1}{2}$ (b) 97 (c) $98\frac{1}{2}$ (d) 98

Q.20. If the following interchanges are made in signs and numbers, which equation would be correct ?
Interchanges: Signs: $\div$ and $-$ Numbers: 15 and 5
SSC CHSL 13/10/2020 (Morning)
(a) $4 \times 30 - 15 \div 5 + 12 = 21$
(b) $12 \times 30 - 15 \div 5 + 4 = 37$
(c) $4 \times 30 - 15 \div 5 + 12 = 27$
(d) $4 \times 30 - 5 \div 15 + 14 = 21$

Q.21. If $\frac{4}{1 + \sqrt{2} + \sqrt{3}} = a + b\sqrt{2} + c\sqrt{3} - d\sqrt{6}$, where a, b, c, d are natural numbers, then the value of $a + b + c + d$ is:
SSC CHSL 18/03/2020 (Morning)
(a) 0 (b) 2 (c) 4 (d) 1

Q.22.

$$\text{If } \left[\left(\frac{2}{3}\right)^3(2x+3)\right]^{\frac{-3}{4}} = \left[\left(\frac{2}{3}\right)^{\frac{2}{3}}(3x+7)\right]^{\frac{-6}{5}}$$

then the value of $\sqrt{2 - 42x}$ is:
SSC CHSL 18/03/2020(Morning)
(a) 5 (b) 6 (c) 3 (d) 4

Q.23. If '+' means '-', '-' means '+', 'x' means '÷' and '÷' means 'x' then the value of $\frac{42 - 12 \times 3 + 8 \div 2 + 15}{8 \times 2 - 4 + 9 \div 3}$ is:

SSC CGL 03/03/2020 (Morning)

(a) $-\frac{5}{3}$ (b) $-\frac{15}{19}$ (c) $\frac{15}{19}$ (d) $\frac{5}{3}$

Q.24. The value of

$$\frac{(4.6)^4 + (5.4)^4 + (24.84)^2}{(4.6)^2 + (5.4)^2 + 24.84}$$
 is:

SSC CGL Tier II (13/09/2019)

(a) 24.42 (b) 24.24 (c) 25.42 (d) 25.48

Q.25. The value of

$$\frac{(0.545)(0.081)(0.51)(5.2)}{(0.324)^3 + (0.221)^3 - (0.545)^3}$$
 is:

SSC CGL Tier II (12/09/2019)

(a) -1 (b) 1 (c) 3 (d) -3

Q.26. The value of $22.\bar{4} + 11.5\bar{67} - 33.\bar{59}$ is
SSC CGL Tier II (11/09/2019)

(a) $0.\bar{32}$ (b) $0.\bar{412}$ (c) $0.3\bar{1}$ (d) $0.4\bar{12}$

Q.27. Find the value

$$\left\{ \frac{(0.7)^2 \div 0.14 + (0.6)^2 \div 0.18 + (0.5)^2 \div 0.05}{4(2.5 \text{ of } 4 - 13 \times 0.25 \times 3)} \right\}$$

SSC MTS 07/08/2019 (Morning)

(a) $\frac{25}{2}$ (b) $\frac{19}{2}$ (c) $\frac{23}{2}$ (d) $\frac{21}{2}$

Q.28. If $x = \frac{1}{12.13} + \frac{1}{13.14} + \frac{1}{14.15}$

.....+ $\frac{1}{23.24}$, $y = \frac{1}{36.37} + \frac{1}{37.38} +$

$\frac{1}{38.39}$ + $\frac{1}{71.72}$ then $\frac{x}{y}$ is equal to:

SSC CHSL 10/07/2019 (Evening)

(a) $\frac{1}{3}$ (b) $\frac{1}{24}$ (c) $\frac{1}{72}$ (d) 3

Q.29. The simplified value of

$$\frac{0.01404}{24^2 + 6^2 - 144}$$
 is:

SSC CHSL 10/07/2019 (Evening)

(a) 3×10^{-5} (b) 6×10^{-5}

(c) 2.4×10^{-4} (d) 3×10^{-4}

Q.30. The simplified value of

$$\frac{1.0025 + 6.25 \times 10^{-6}}{0.0025 + 0.95}$$
 is:

SSC CHSL 09/06/2019 (Evening)

(a) 1.0025 (b) 1.0525

(c) 1.0005 (d) 1.0505

Practice Questions

SSC CHSL 2024 Tier - 2

Q.31. The value of $5 + 24 \times 8 \div 8$ of $8 - 8 \div 8 \times 4$ of 4 is equal to:

SSC CHSL Tier II (18/11/2024)

(a) 8 (b) 9 (c) -8 (d) -9

Q.32. Simplify $(2z - 5y)^2 + (5z + 2y)^2 - 25z^2$.

SSC CHSL Tier II (18/11/2024)

(a) $19y^2 - 4z^2$ (b) $29y^2 + 4z^2$

(c) $29y^2 - 4z^2$ (d) $19y^2 + 4z^2$

SSC MTS 2024 Tier - 1

Q.33. The lowest terms of $\frac{221}{488}$ is:

SSC MTS 30/10/2024 (Afternoon)

(a) $\frac{121}{488}$ (b) $\frac{221}{488}$ (c) $\frac{121}{288}$ (d) $\frac{221}{288}$

SSC CGL 2024 Tier - 1

Q.34. Simplify $\frac{(x^2 - 9)}{x + 3}$

SSC CGL 09/09/2024 (1st Shift)

(a) $x + 3$ (b) $x - 3$ (c) $\sqrt{x^2 - 9}$ (d) $x - 9$

Q.35. Simplify

$$\frac{(8.3)^3 + (9.2)^3 + (6.1)^3 - 3 \times 8.3 \times 9.2 \times 6.1}{(8.3)^2 + (9.2)^2 + (6.1)^2 - 8.3 \times 9.2 - 9.2 \times 6.1 - 6.1 \times 8.3}$$

SSC CGL 09/09/2024 (2nd Shift)

(a) 30.2 (b) 23.6 (c) 28.7 (d) 25.5

Q.36. Simplify $\frac{0.58 \times 0.58 \times 0.58 - 0.001}{0.58 \times 0.58 + 0.058 + 0.01}$

SSC CGL 10/09/2024 (2nd Shift)

(a) 0.48 (b) 0.28 (c) 0.37 (d) 0.42

Q.37. If $45 \div 3 + 4 \times A - 48 \div 24 + 4 = 1$, then the value of A is:

SSC CGL 10/09/2024 (3rd Shift)

(a) -2 (b) 4 (c) 2 (d) -4

Q.38. Simplify $\frac{(4.2)^3 - (1.2)^3}{(4.2)^2 + 5.04 + (1.2)^2}$

SSC CGL 11/09/2024 (2nd Shift)

(a) 2 (b) 3 (c) 1 (d) 5

Q.39. The value of the expression $(1.55^2 + 0.55^2 + 1.55 \times 0.55)$ is equal to:

SSC CGL 11/09/2024 (2nd Shift)

(a) $1.55^2 - 0.55^2$ (b) $1.55^3 - 0.55^3$

(c) $1.55^3 + 0.55^3$ (d) $1.55^2 + 0.55^2$

Q.40. Solve the following:

$$[(2.5 + 1.5)^2 \div 4 \text{ of } \frac{1}{2} \times 8 - 1]$$

SSC CGL 12/09/2024 (2nd Shift)

(a) 0 (b) 81 (c) 27.1 (d) 63

Q.41. Simplify the following expression.

$$\frac{(x-y)^3 + (y-z)^3 + (z-x)^3}{(x-y)(y-z)(z-x)}$$

SSC CGL 13/09/2024 (1st Shift)

(a) 3 (b) 1 (c) 0 (d) 2

Q.42. Simplify the following expression.

SSC CGL 13/09/2024 (3rd Shift)

$$(g-h)^2 - (g+h)^2 + 4gh$$

(a) 4gh (b) -4gh (c) 4 (d) 0

Q.43. Simplify

$$\frac{1.5 \times 1.5 + 2.5 \times 2.5 + 3.5 \times 3.5 + 2 \times 1.5 \times 2.5 + 2 \times 2.5 \times 3.5 + 2 \times 1.5 \times 3.5}{1.5 + 2.5 + 3.5}$$

SSC CGL 13/09/2024 (3rd Shift)

(a) 9.5 (b) 6.5 (c) 8.5 (d) 7.5

Q.44. Simplify the following:

$$[(5\frac{1}{2} - 3\frac{1}{3}) \text{ of } 1\frac{5}{13} + 1 - 2 \times 3\frac{5}{2} \div 2\frac{3}{4}]$$

SSC CGL 18/09/2024 (1st Shift)

(a) 0 (b) $\frac{11}{2}$ (c) -2 (d) $3\frac{1}{2}$

Q.45. Simplify the given expression.

$$\frac{(4x^2 + 9y^2 - 12xy)}{(3y - 2x)}$$

SSC CGL 18/09/2024 (3rd Shift)

(a) $2y - 3x$ (b) $3y - 2x$ (c) $2x - 3y$ (d) $3x - 2y$

Q.46. Simplify the following expression
 $25 \div 5 + 3$ of $2 - 4 + 6$ of $3 \div 2 - 8 + 4$

SSC CGL 19/09/2024 (2nd Shift)

(a) 16 (b) 4 (c) 8 (d) 12

Q.47. Simplify the following expression.
 $0.07 \times 0.28 \div 0.04 + 0.64 - 1.64 \div 0.04$

SSC CGL 23/09/2024 (3rd Shift)

(a) 38.56 (b) -39.87 (c) -46.48 (d) 36.78

Q.48. Find the value of

$$11 - \{(3 \div 4) \text{ of } 8 - 4 - 8\}.$$

SSC CGL 24/09/2024 (1st Shift)

(a) 2 (b) 3 (c) 1 (d) -1

Q.49. Simplify the following expression.
 $133 \div 19 \times 3 - 48 \times 2 \div 24 + 841 \div$

$$[45 - 16]$$

SSC CGL 24/09/2024 (2nd Shift)

(a) 46 (b) 21 (c) 12 (d) 42

Q.50. $9 \div [1 + \{4 \times (\frac{5}{6} - \frac{1}{3} + \frac{1}{2})\}]$

is equal to:

SSC CGL 25/09/2024 (1st Shift)

(a) 1 (b) $\frac{9}{5}$ (c) 0 (d) 9

Q.51. The value of N in the equation

$$9 \times N \text{ of } 4 \div 4 - 15 \times 4 = 12$$
 is:

SSC CGL 25/09/2024 (3rd Shift)

(a) 9 (b) 0 (c) 12 (d) 8

Q.52. If $\frac{3(16^3 - 6^3)}{16^2 + 6^2 + Q} = 30$, then find the value of Q.

SSC CGL 26/09/2024 (1st Shift)

(a) 96 (b) 98 (c) 112 (d) 108

Q.53. Simplify the following expression.
 $(60 + 64 \div 4 \text{ of } 4) \times \{1000 \div ((10 \text{ of } (3 + 2)) \times 2 \text{ of } 5)\} - 28 + (100 \times 3 \text{ of } 3)$.
 SSC CGL 26/09/2024 (2nd Shift)
 (a) 600 (b) 1000 (c) 100 (d) 500

Q.54. Simplify $626 \times 2912 \div 8138$
 SSC CGL 26/09/2024 (3rd Shift)
 (a) 285 (b) 224 (c) 364 (d) 310

Q.55. Simplify $\frac{4x^2 - 49y^2}{2x + 7y} + 7y - 2x$
 SSC CGL 26/09/2024 (3rd Shift)
 (a) $4x - 14y$ (b) $7y$ (c) $4x$ (d) 0

SSC CHSL 2024 Tier - 1

Q.56. The value of $\left(\frac{7 + 8 \times 7 \div 8 + 7 \div 7 \div 7}{3 + 3 \times 3 \div 3 + 5 \div 5 \div 5}\right)$ is :
 SSC CHSL 01/07/2024 (2nd Shift)
 (a) $1 \frac{81}{217}$ (b) $2 \frac{61}{217}$
 (c) $3 \frac{27}{217}$ (d) $1 \frac{31}{217}$

Q.57. What is the value of the following expression ?
 $\frac{4}{7} \times 4 \frac{1}{2} \div 5 \frac{1}{3} \text{ of } 2 \frac{1}{2} - (7 \frac{7}{8} \div 5 \frac{1}{9} \text{ of } 9 \frac{9}{20}) + \frac{1}{2}$
 SSC CHSL 01/07/2024 (2nd Shift)
 (a) $\frac{1759}{3250}$ (b) $\frac{1113}{856}$ (c) $\frac{853}{1610}$ (d) $\frac{1093}{111}$

Q.58. Simplify the following expression.
 $(3 - 3 \times 3 + 3 \div 3 + 3 \times 5) \times 2 \text{ of } 5 + (2 + 2 \div 2 + 2 \times 2 - 2)$
 SSC CHSL 02/07/2024 (2nd Shift)
 (a) 102 (b) 100 (c) 105 (d) 106

Q.59. Simplify:
 $\{(3^3 + 2^3) \times \frac{22}{7}\} \times \{(6^2 + 3^2 - 4^2) \div (14 - \frac{1}{2})\}$
 SSC CHSL 02/07/2024 (3rd Shift)
 (a) 220 (b) 145 (c) 110 (d) 198

Q.60. Simplify the following expression.
 $[40 - 48 \div 4 \text{ of } 3] \div \{(6 + 4) - 4 \text{ of } 4 + 4 + (6 - 4) + (4 \times 4) \div 4\} \times 3$
 SSC CHSL 02/07/2024 (3rd Shift)
 (a) 30 (b) 26 (c) 25 (d) 27

Q.61. Simplify the given expression.
 $8 - 5 \div 2^3 \times 3 + (7 \div 7 \times 7) \div 7 + (3 \times 6 \div \frac{1}{3}) \times \frac{1}{3^3}$
 SSC CHSL 03/07/2024 (3rd Shift)
 (a) $\frac{73}{9}$ (b) $\frac{73}{27}$ (c) $\frac{73}{21}$ (d) $\frac{73}{8}$

Q.62. The value of $10 + 3 \frac{3}{5} \div [2 \frac{2}{3} \div \{7 \frac{1}{6} - (2 \frac{1}{2} + \frac{2}{3})\}]$ is equal to :

SSC CHSL 04/07/2024 (2nd Shift)
 (a) $\frac{27}{55}$ (b) $\frac{77}{5}$ (c) $\frac{55}{27}$ (d) $\frac{5}{77}$

Q.63. Simplify:
 $3 \frac{4}{7} - [16 \frac{1}{2} \div \{6^2 - (7 - 2 \frac{1}{2})\}]$
 SSC CHSL 04/07/2024 (3rd Shift)
 (a) $\frac{25}{11}$ (b) $\frac{64}{21}$ (c) $\frac{64}{11}$ (d) $\frac{25}{21}$
Q.64. Simplify $\frac{\frac{1}{2} \div \frac{3}{4} \text{ of } \frac{5}{6}}{\frac{1}{2} \div \frac{3}{4} \times \frac{5}{6}} + \frac{2 + 2 \times 2}{2 \div 2 \times 2}$

SSC CHSL 05/07/2024 (2nd Shift)
 (a) $\frac{111}{25}$ (b) $\frac{121}{25}$ (c) $\frac{75}{36}$ (d) 4

Q.65. Simplify the following expression.
 $4 \times 7 - 3\{7 \times 4 \div (6 \times 2)\}$
 SSC CHSL 08/07/2024 (1st Shift)
 (a) 12 (b) 19 (c) 21 (d) 4

Q.66. What is the value of the following expression?
 $420 \div [\frac{4}{9} \times (1 + \frac{4}{5}) \times 5] + [\frac{5}{7} \text{ of } \frac{21}{25} \div \frac{3}{4} \text{ of } \frac{2}{5}]$

SSC CHSL 08/07/2024 (3rd Shift)
 (a) 127 (b) 139 (c) 107 (d) 89
Q.67. Solve $\frac{2 \times 2 + 2}{2 \div 2 \times 2} \div \frac{\frac{1}{2} \div \frac{1}{2} \text{ of } \frac{1}{2}}{\frac{1}{2} + \frac{1}{2} \text{ of } \frac{1}{2}}$

SSC CHSL 08/07/2024 (4th Shift)
 (a) $4 \frac{1}{2}$ (b) $2 \frac{1}{8}$ (c) $1 \frac{1}{8}$ (d) $3 \frac{1}{8}$

Q.68. What is the value of the following expression ?
 $(24 \div 3 \text{ of } \frac{1}{6}) \times (\frac{4}{5} \div \frac{9}{8} \times \frac{3}{5}) \div (\frac{4}{7} \div \frac{3}{14} \text{ of } \frac{3}{14})$

SSC CHSL 09/07/2024 (2nd Shift)
 (a) $\frac{379}{179}$ (b) $\frac{579}{459}$ (c) $\frac{363}{553}$ (d) $\frac{288}{175}$

Q.69. The value of $\frac{5 + 7 \times 5 \div 8 \text{ of } 5 + 3 \div 3 \times 5}{6 \div 6 \text{ of } 4 + 9 \times 9 \div 3 \times 2 \times 2}$ is :
 SSC CHSL 09/07/2024 (3rd Shift)
 (a) $\frac{67}{859}$ (b) $\frac{87}{866}$ (c) $\frac{97}{881}$ (d) $\frac{77}{861}$

Q.70. Simplify the following expression.
 $11 \frac{1}{4} \div 3 \frac{3}{5} \times 6 \frac{2}{3} + \frac{1}{6} = \underline{\hspace{2cm}}$
 SSC CHSL 09/07/2024 (4th Shift)
 (a) 19 (b) 21 (c) 23 (d) 25

Q.71. The value of $[3^3 \{(40 \times 5) + (7 \times 6)\} \div 11] + (2^3 \times 15 \div 24)$ is:
 SSC CHSL 10/07/2024 (3rd Shift)
 (a) 559 (b) 595 (c) 599 (d) 959

Q.72. If '-' means 'x', 'x' means '÷', '+'

means '-', '-' means '+', then $47 \div (10 \times 5) + (4 - 5) - 6$ is:

SSC CHSL 10/07/2024 (4th Shift)
 (a) 100 (b) 59 (c) 79 (d) 91

Q.73. What is the value of $\frac{[\frac{1}{4} \div \frac{1}{4} \times \frac{1}{4}] \times \frac{1}{4}}{[\frac{1}{8} + \frac{1}{8} \times \frac{1}{8}]} + (1 - \frac{4}{9})$?

SSC CHSL 11/07/2024 (1st Shift)
 (a) 1 (b) 0 (c) 3 (d) 2

SSC CPO 2024 Tier - 1

Q.74. Simplify the following expression.
 $3 \frac{1}{5} \text{ of } 2 \frac{3}{4} \div \frac{11}{3} + (2 \frac{1}{5} \div 1 \frac{2}{5} \text{ of } 1 \frac{3}{7} - 2 \frac{1}{3})$

SSC CPO 27/06/2024 (1st Shift)
 (a) $4 \frac{1}{6}$ (b) $2 \frac{1}{6}$ (c) $3 \frac{1}{6}$ (d) $1 \frac{1}{6}$

Q.75. Find the value of $[18 - 3 \times \{3 - 3 \times (3 \div \frac{1}{3} - 3) + 3 \div \frac{1}{3} + 3\} - 4 \times 2]$.
 SSC CPO 27/06/2024 (1st Shift)
 (a) 18 (b) 19 (c) 12 (d) 15

Q.76. Find the value of $12 + 24 \div 2 - 7 \times 30 \div 6 + (5 + 4) \times 8 + 8 \div 8 - 8$.
 SSC CPO 27/06/2024 (1st Shift)
 (a) 45 (b) 85 (c) 54 (d) 58

Q.77. The possible value of x satisfying $\frac{(143 \div 13 \times x - 3 \times 3)}{(6^2 - 7 \times 5 + x^2)} = 1$ is :
 SSC CPO 27/06/2024 (1st Shift)
 (a) 1 (b) 3 (c) 2 (d) 4

Q.78. Find the value of $\frac{12 + 3 \frac{2}{3} \div [1 \frac{3}{4} \div \{5 \frac{1}{4} \times (3 \frac{3}{5} \div 4 \frac{1}{2})\}]}{1 \frac{1}{3} \times 5 \frac{1}{5}}$

SSC CPO 27/06/2024 (2nd Shift)
 (a) 5 (b) 3 (c) 1 (d) 2

Q.79. Simplify the following expression:
 $25 \times 25 + 9 \times 9 + 6 \times 6 + 2 \times 25 \times 9 + 2 \times 9 \times 6 + 2 \times 25 \times 6$
 SSC CPO 27/06/2024 (2nd Shift)
 (a) 900 (b) 1200 (c) 1600 (d) 2500

Q.80. The value of $1 - \frac{7}{1 - \frac{1}{1 - \frac{7}{1 - 7}}}$ is :
 SSC CPO 27/06/2024 (2nd Shift)
 (a) 2 (b) 0 (c) 1 (d) -1

Q.81. Simplify $9^{18} \div 3^{14} \text{ of } 27^3 \times \sqrt{6561}$
 SSC CPO 27/06/2024 (2nd Shift)
 (a) 3^{17} (b) 3^{16} (c) 3^{15} (d) 3^{18}

Q.82. The value of $3 + 2 \times [7 + (5 + 7) \div 2]$ is equal to :

SSC CPO 27/06/2024 (2nd Shift)

(a) 39 (b) 65 (c) 29 (d) 47

Q.83. Simplify $\frac{92736}{(274 + 830) \text{ of } \frac{7}{4}} + 513$

SSC CPO 27/06/2024 (3rd Shift)

(a) 561 (b) 526 (c) 546 (d) 574

Q.84. Simplify the following expression:

$$\frac{(525 + 175)^2 - (525 - 175)^2}{(525 \times 175)}$$

SSC CPO 27/06/2024 (3rd Shift)

(a) 350 (b) 700 (c) 2 (d) 4

Q.85. The value of $9 \times 9 + 9 \div 9 - 9 \times 9 + 9 + 9 \times 9 - 9 - 9 \times 9$ is:

SSC CPO 27/06/2024 (3rd Shift)

(a) 9 (b) 1 (c) 81 (d) 0

Q.86. The value of $(4^2)^2 + 4 \times 4 \div 4 - 4$ is:

SSC CPO 27/06/2024 (3rd Shift)

(a) 4^4 (b) 4^3 (c) 4^2 (d) 4^0

Q.87. The value of $1 + \frac{1}{1 + \frac{1}{5 + \frac{1}{2 + \frac{1}{3}}}}$ is:

SSC CPO 27/06/2024 (3rd Shift)

(a) $1\frac{13}{45}$ (b) $2\frac{17}{45}$ (c) $2\frac{11}{45}$ (d) $1\frac{38}{45}$ **Q.88.** If $x [-5 \{-4(-a)\}] + 6 [-3 \{-3(-a)\}] = 6a$, then the value of x is:

SSC CPO 28/06/2024 (1st Shift)

(a) -1 (b) 0 (c) -3 (d) 2

Q.89. Simplify $20 \div 5$ of $4 \times 8 + 6 \div 24$ of $3 \times 48 - 6 \div 18 \times 9$.

SSC CPO 28/06/2024 (1st Shift)

(a) 9 (b) 6 (c) 4 (d) 1.125

Q.90. The value of $64 \div [44 - (8 \text{ of } 3 - 16) \div 4 \times 20]$ is equal to:

SSC CPO 28/06/2024 (1st Shift)

(a) 12 (b) 18 (c) 16 (d) 15

Q.91. If $855 \div 15 - k + 32 \times 5 = 1152 \div 16 \times 111 \div 37$, then the value of k is:

SSC CPO 28/06/2024 (2nd Shift)

(a) 2 (b) 1 (c) 0 (d) 3

Q.92. Simplify the given expression.

$$1000 \times 100 - 1000 + 4000 \div 200$$

SSC CPO 28/06/2024 (2nd Shift)

(a) 99235 (b) 99980 (c) 99020 (d) 99999

Q.93. Find the value of 15% of $1.5 + 0.15 \div 0.1 + 0.01$ of $1.1 \times 0.5 + 2.5 \times 25$.

SSC CPO 28/06/2024 (2nd Shift)

(a) 77.2305 (b) 64.2305

(c) 62.2305 (d) 6.42305

Q.94. Simplify the following expression.

$$(6 + 6 \text{ of } 5 \div 5) \div 6 \text{ of } 5 + (24 \div 2 \text{ of } 3 - 6 \text{ of } 5 + 4)$$

SSC CPO 28/06/2024 (2nd Shift)

(a) $-26\frac{2}{5}$ (b) $-17\frac{2}{5}$ (c) $-21\frac{3}{5}$ (d) $-23\frac{1}{5}$ **Q.95.** Find the simplified value of $5055 - (1002 \div 20.04)$.

SSC CPO 28/06/2024 (2nd Shift)

(a) 4050 (b) 5005 (c) 50.05 (d) 5050

Q.96. If $56 \div 4 + p \times 3 - 24 \div 12 + 4 = 21 \div 3 + 3$, then the value of p is:

SSC CPO 28/06/2024 (3rd Shift)

(a) 2 (b) -1 (c) 1 (d) -2

Q.97. Simplify

$$22.0625 - [15 + 0.25 \times (4.78 - 1.5 \times 3.02)]$$

SSC CPO 28/06/2024 (3rd Shift)

(a) 7.3 (b) 7.0 (c) 7.1 (d) 7.2

Q.98. Simplify the following expression.

$$(4.2 \text{ of } 2.5 \div 3.5) - (3.6 \div 2.4 + 2.5 \text{ of } 3 \div 7.5) + (6 - 3 \text{ of } 2 + 4 \text{ of } 3 \div 6)$$

SSC CPO 28/06/2024 (3rd Shift)

(a) 2.5 (b) 4.5 (c) 3.5 (d) 1.5

Q.99. Simplify the given expression.

$$\frac{13 \div \{6 - 6 \div (7 - 9) \times 8 + 9\}}{5 + 5 \times 5 \div 5 \text{ of } 5}$$

SSC CPO 28/06/2024 (3rd Shift)

(a) 13 (b) 39 (c) $\frac{1}{18}$ (d) $\frac{1}{2}$ **Q.100.** Simplify the given expression.

$$\frac{(2\frac{1}{7} + 6\frac{1}{7}) \div \frac{9}{7}}{1\frac{1}{9} \div \{\frac{7}{9} + (\frac{4}{9} \div \frac{4}{3})\}}$$

SSC CPO 29/06/2024 (1st Shift)

(a) $\frac{53}{9}$ (b) $\frac{56}{9}$ (c) $\frac{58}{9}$ (d) $\frac{55}{9}$ **Q.101.** The value of

$$\frac{(275 \times 275 \times 275 + 725 \times 725 \times 725)}{(275 \times 275 + 725 \times 725 - 725 \times 275)} \times \frac{(275 \times 275 \times 275 - 725 \times 725 \times 725)}{(275 \times 275 + 725 \times 725 + 725 \times 275)}$$

SSC CPO 29/06/2024 (2nd Shift)

(a) -450000 (b) 45000

(c) 450000 (d) -45000

Q.102. If $\frac{5x + 3y}{5x - 3y} = \frac{7}{3}$, then find $x : y$

SSC CPO 29/06/2024 (2nd Shift)

(a) 5 : 3 (b) 3 : 2 (c) 3 : 5 (d) 2 : 3

Q.103. The value of $\frac{1}{2^2} + \frac{1}{2^3} \div (\frac{1}{2} + \frac{1}{2} \div 1) + 3\frac{1}{3} \div \frac{5}{2} \times \frac{3}{4} \div 6\frac{2}{3} \times \frac{7}{6}$ is:

SSC CPO 29/06/2024 (2nd Shift)

(a) $\frac{11}{20}$ (b) $\frac{11}{30}$ (c) $\frac{21}{20}$ (d) $\frac{7}{40}$ **Q.104.** Simplify the given expression.

$$12.3 - [7 + 0.8 \text{ of } (5.6 - 3.8 \times 1.03)]$$

SSC CPO 29/06/2024 (2nd Shift)

(a) 5.2314 (b) 7.8539

(c) 1.2368 (d) 3.9512

Q.105. If $x + \frac{1}{1 + \frac{1}{3 + \frac{1}{2}}} = 1$, then thevalue of x is equal to:

SSC CPO 29/06/2024 (2nd Shift)

(a) $\frac{7}{9}$ (b) $\frac{1}{7}$ (c) $\frac{3}{7}$ (d) $\frac{2}{9}$ **Q.106.** Simplify the given expression.

$$55\frac{1}{3} \div [3\frac{1}{5} + \{2\frac{1}{2} - (2\frac{1}{3} + \frac{3}{5})\}]$$

SSC CPO 29/06/2024 (3rd Shift)

(a) 25 (b) 35 (c) 20 (d) 15

Q.107. Simplify

$$8.16 \times 5.35 + 17.9 - 19.5 \text{ of } \frac{1}{15} + \frac{1}{5}$$

SSC CPO 29/06/2024 (3rd Shift)

(a) 57.426 (b) 60.456

(c) 65.234 (d) 70.225

Q.108. Simplify the given expression.

$$\frac{53 - \frac{2}{7} \text{ of } 63 - 9}{72 - \frac{2}{5} \text{ of } (96 - 31)}$$

SSC CPO 29/06/2024 (3rd Shift)

(a) $\frac{26}{23}$ (b) $\frac{11}{23}$ (c) $\frac{13}{23}$ (d) $\frac{7}{23}$ **SSC Selection Post (Phase - XII)****Q.109.** Simplify: $25^3 - 75^3 + 50^3$

Higher Secondary 20/06/2024 (Shift - 2)

(a) 271250 (b) -281450

(c) 281350 (d) -281250

Q.110. Simplify the expression:

$$(0.2)^3 + (0.3)^3 + (0.5)^3 + 3 \times 0.5 \times 0.7 \times 0.8$$

Higher Secondary 26/06/2024 (Shift - 2)

(a) 0.001 (b) 1.525 (c) 1.000 (d) 0.005

Q.111. Simplify the following expression:

$$\frac{0.7^2 + (0.4 + 0.2)(0.7) + (0.4 \times 0.2)}{(0.7 + 0.4)}$$

Graduate Level 20/06/2024 (Shift - 3)

(a) 0.9 (b) 0.7 (c) 0.3 (d) 0.4

Q.112. Simplify the following expression

$$\frac{(8.25)^3 - 3 \times (8.25)^2 \times 0.25 + 24.75 \times (0.25)^2 - (0.25)^3}{4 \times 4 \times 4}$$

Graduate Level 21/06/2024 (Shift - 3)

(a) 6 (b) 4 (c) 2 (d) 8

Q.113. Simplify the given expression

$$\frac{(2.3)^3 + (15.9)^3 + (3.7)^3 - 3 \times 2.3 \times 15.9 \times 3.7}{(2.3)^2 + (15.9)^2 + (3.7)^2 - 2.3 \times 15.9 - 15.9 \times 3.7 - 3.7 \times 2.3}$$

Graduate Level 21/06/2024 (Shift - 3)

(a) 14.5 (b) 20.9 (c) 17.6 (d) 21.9

Q.114. Simplify the given expression.

$$\frac{12.43 \times 12.43 \times 12.43 + 19.57 \times 19.57 \times 19.57}{12.43 \times 12.43 - 12.43 \times 19.57 + 19.57 \times 19.57}$$

Graduate Level 24/06/2024 (Shift - 3)

(a) 36 (b) 30 (c) 32 (d) 34

Q.115. Simplify the expression:

$$15 \times 15 \times 15 + 3 \times 15 \times 12 \times 12 - 3 \times 15 \times 15 \times 12 - 12 \times 12 \times 12$$

Graduate Level 25/06/2024 (Shift - 3)

(a) 36 (b) 9 (c) 18 (d) 27

Q.116. Simplify the following expression:

$$\frac{((32)^4 - 1)}{33 \times 31 \times (2^{10} + 1)}$$

Graduate Level 26/06/2024 (Shift - 4)

(a) 2 (b) 1 (c) -2 (d) -1

SSC CHSL 2023 Tier - 2

Q.117. If $\sqrt[3]{0.08 \times 0.8 \times p} = 0.008 \times 0.8 \times \sqrt[3]{q}$, then find the value of $\frac{p}{q}$.

SSC CHSL Tier II (02/11/2023)

(a) $8^3 \times 10^{-8}$ (b) $8^3 \times 10^{-9}$
(c) $8^4 \times 10^{-9}$ (d) $8^4 \times 10^{-8}$

Q.118. Simplify :

$$\left(\sqrt[3]{\sqrt[3]{343}}\right)^3 \times \left(\sqrt[3]{\sqrt[3]{216}}\right)^3$$

SSC CHSL Tier II (10/01/2024)

(a) 56 (b) 42 (c) $\sqrt[3]{42}$ (d) 42^3 **SSC CGL 2023 Tier - 2**

Q.119. Find the value of $\frac{\sqrt{1.24} \times \sqrt{2.79}}{\sqrt{2.64} \times \sqrt{5.94}}$.

SSC CGL Tier II (26/10/2023)

(a) $\frac{31}{66}$ (b) $\frac{31}{44}$ (c) $\frac{33}{64}$ (d) $\frac{33}{31}$ **SSC CPO 2023 Tier - 1****Q.120.** Evaluate

$$\frac{0.04}{0.05} \text{ of } \left(3\frac{1}{3} - 2\frac{1}{2}\right) \div \frac{1}{2} \text{ of } 1\frac{1}{4}$$

SSC CPO 03/10/2023 (1st Shift)

(a) 5 (b) 0.3 (c) 3 (d) 0.03

Q.121. If $(48 \div 72 \times 3) - [15 \div 8 \times (40 - 32) - 10] + 2P = 6 \div 2$, then find the value of P

SSC CPO 03/10/2023 (3rd Shift)

(a) 2 (b) 4 (c) 1 (d) 3

Q.122. The value of

$$\frac{1\frac{1}{2} \div 3\frac{1}{4} + \frac{1}{2} \div \frac{13}{14} + \frac{1}{5}}{\frac{1}{5} \times 3\frac{1}{2} - \frac{1}{3} \div 1\frac{3}{4} \times 3\frac{1}{2}} \text{ is :}$$

SSC CPO 05/10/2023 (1st Shift)

(a) 38 (b) 33 (c) 36 (d) 40

Q.123. Simplify the given expression.

$$b + 2a - [(3b + a) - (2a + b) + 2a] - a$$

SSC CPO 05/10/2023 (3rd Shift)

(a) -a (b) -b (c) a (d) b

SSC MTS 2023 Tier - 1**Q.124.** Simplify the given expression

$$71 - (-3)\{-2 - (8 - 3)\} + 3\{5 + (-2)(-1)\}$$

SSC MTS 04/09/2023 (3rd Shift)

(a) 30 (b) 119 (c) 50 (d) 71

Q.125. Evaluate $3\frac{1}{2} - [2\frac{1}{4} \div \{1\frac{1}{4} - \frac{3}{4}(1\frac{1}{2} - \frac{1}{2} - 2\frac{1}{2})\}]$

SSC MTS 11/09/2023 (1st Shift)

(a) $1\frac{21}{38}$ (b) $2\frac{19}{18}$ (c) $2\frac{21}{38}$ (d) $\frac{47}{38}$ **Q.126.** What is the value of

$$\left[\frac{1}{5} \div \frac{1}{5} \times \frac{1}{5}\right] \times \frac{1}{5} \div \left[\frac{1}{9} + \frac{1}{9} \times \frac{1}{9}\right] \times \frac{1}{10} + \left(1 - \frac{6}{25}\right)$$

SSC MTS 12/09/2023 (2nd Shift)

(a) 2 (b) 1 (c) 3 (d) 4

SSC CHSL 2023 Tier - 1

Q.127. If the mathematical symbols in the expression given below are changed from '+' to '÷', '-' to '×', '÷' to '-' and '×' to '+', then what will be the value of the expression?

$$87 \times 256 + 16 - 77 \div 1359$$

SSC CHSL 07/08/2023 (3rd Shift)

(a) 20 (b) -40 (c) 40 (d) -20

Q.128. If $\sqrt{a} = 3b$, then $\frac{a}{b^2}$ is equal to :

SSC CHSL 08/08/2023 (2nd Shift)

(a) $\frac{1}{6}$ (b) 6 (c) $\frac{1}{9}$ (d) 9**Q.129.** Simplify the given expression.

$$\frac{(0.25)^4 + 2 \times (0.25)^2 + 1 - (0.25)^2}{(0.25)^2 + 0.25 + 1}$$

SSC CHSL 08/08/2023 (2nd Shift)

(a) 0.6755 (b) 0.9025
(c) 0.8125 (d) 0.7835

Q.130. Evaluate $\frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}}$, given that

$$\sqrt{6} = 2.45$$

SSC CHSL 08/08/2023 (4th Shift)

(a) 7.7 (b) 9.9 (c) 8.8 (d) 6.6

Q.131. Simplify the given expression

$$\frac{(4.2)^3 - 0.008}{(4.2)^2 + 0.84 + 0.04}$$

SSC CHSL 14/08/2023 (2nd Shift)

(a) 4 (b) -2 (c) 2 (d) -4

Q.132. Simplify the given expression and find the value for x = -1.

$$\frac{10x^2 + 5x + 2xy + y}{5x + y}$$

SSC CHSL 17/08/2023 (3rd Shift)

(a) -1 (b) 0 (c) 1 (d) 2

SSC CGL 2023 Tier - 1**Q.133.** Simplify the given expression.

$$y + 2x - [y - (y - x + y) - (x + y) + y] - 2y$$

SSC CGL 17/07/2023 (1st Shift)

(a) -y (b) -2x (c) y (d) 2x

Q.134. Simplify the following

$$\frac{3a + b}{2} - \frac{a - 3b}{3} + 2b$$

SSC CGL 19/07/2023 (1st shift)

(a) $\frac{5(a + 3b)}{6}$ (b) $\frac{7a + 3b}{6}$
(c) $\frac{7(a + 3b)}{6}$ (d) $\frac{a + 3b}{6}$

Q.135. Simplify the given expression.

$$(a - 2b)(b - 3a) + (a + b)(a - 3b) - (b - 3a)(4a - 5b)$$

SSC CGL 21/07/2023 (1st shift)

(a) $8a^2 - 10ab$
(b) $8a^2 + 10ab$
(c) $10a^2 - 14ab$
(d) $10a^2 - 4ab$

Q.136. Simplify the following expression.

$$(4x + 1)^2 - (4x + 3)(4x - 1)$$

SSC CGL 24/07/2023 (2nd shift)

(a) $(4x + 1)$ (b) 4 (c) $(4x - 3)$ (d) $4x$

Q.137. What value of 'y' will satisfy the following equation?

$$\frac{3y}{1 + \frac{1}{1 + \frac{y}{1 - y}}} = 2$$

SSC CGL 25/07/2023 (1st shift)

(a) $\frac{2}{3}$ (b) $\frac{5}{4}$
(c) $\frac{4}{5}$ (d) $\frac{3}{2}$

Q.138. Simplify the expression $\frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}}$,

where x = 2 and y = 3.

SSC CGL 25/07/2023 (4th shift)

(a) $2\sqrt{6} - 6$ (b) $\sqrt{6} - 5$
(c) $5 - 2\sqrt{6}$ (d) $2\sqrt{6} - 5$

SSC Selection Post (Phase - XI)**Q.139.** Simplify $52 \times 52 - 48 \times 48$.

Higher Secondary 27/06/2023 (Shift - 3)

(a) 386 (b) 400 (c) 424 (d) 326

SSC CHSL 2022 Tier - 2

Q.140. What is the correct descending order of the following ratios?

$$\frac{17}{30}, \frac{7}{15}, \frac{27}{50}, \frac{11}{20}$$

SSC CHSL Tier II (26/06/2023)

(a) $\frac{11}{20} > \frac{17}{30} > \frac{27}{50} > \frac{7}{15}$
(b) $\frac{17}{30} > \frac{11}{20} > \frac{27}{50} > \frac{7}{15}$
(c) $\frac{17}{30} > \frac{27}{50} > \frac{11}{20} > \frac{7}{15}$
(d) $\frac{7}{15} > \frac{11}{20} > \frac{27}{50} > \frac{17}{30}$

SSC MTS 2022 Tier - 1

Q.141. The value of

$$\left(\frac{1}{5}\right) + \left[1\frac{1}{4} - \frac{1}{2}\right] \times \left(\frac{2}{5}\right) \text{ lies between :}$$

SSC MTS 04/05/2023 (Morning)

- (a) 2.1 and 2.4 (b) 0.2 and 0.6
(c) 1.5 and 1.8 (d) 0.8 and 1.2

Q.142. Find the value of $0.5 + 0.55 + 0.555 + 0.5555$:

SSC MTS 09/05/2023 (Evening)

- (a) 1.4405 (b) 1.0555
(c) 1.5555 (d) 2.1605

Q.143. Find the value of

$$\sqrt{[(7^2 + 6^3 + 9^2) \div 2 + 52]}$$

SSC MTS 16/05/2023 (Morning)

- (a) 14 (b) 15 (c) 16 (d) 13

Q.144. What is the value of

$$\sqrt{0.52 \times 0.52 + 0.31 \times 0.31 + 0.62 \times 0.52}$$

SSC MTS 13/06/2023 (Morning)

- (a) 0.63 (b) 0.73 (c) 0.83 (d) 0.93

Q.145. Find the value of :

$$\frac{49}{63} \div \frac{21}{9} - \frac{85}{510} + \frac{88}{132} \div \frac{66}{11}$$

$$\frac{1}{7} \div \frac{33}{21} + \frac{26}{143} - \frac{1}{209} \div \frac{1}{19}$$

SSC MTS 19/06/2023 (Afternoon)

- (a) $\frac{55}{36}$ (b) $\frac{55}{39}$ (c) $\frac{13}{46}$ (d) $\frac{44}{13}$

SSC CHSL 2022 Tier - 1

Q.146. What is the value of

$$\frac{5}{2} \div \frac{2}{5} \times \frac{1}{5} - \frac{1}{5} \div \frac{3}{5} \times \frac{5}{2}$$

$$\frac{2}{5} \div \frac{1}{5} - \frac{1}{2} \times \frac{5}{2} \div \frac{25}{4} + \frac{1}{2} \div \frac{2}{5} \times \frac{1}{5}$$

SSC CHSL 09/03/2023 (1st Shift)

- (a) $\frac{31}{127}$ (b) $\frac{32}{127}$ (c) $\frac{28}{125}$ (d) $\frac{25}{123}$

Q.147. If $A = \frac{(36 \div 18 \times 5)}{(36 \div (3 \times 4))}$ and
 $B = \frac{(5 \div 15 \times 5 + 6 \times 2 - 5)}{(15 \div 5 \times 25 - 2 \times 3 + 6)}$ then what is the value of $(A + B)$?

SSC CHSL 09/03/2023 (4th Shift)

- (a) $\frac{776}{225}$ (b) $\frac{424}{225}$ (c) $\frac{772}{225}$ (d) $\frac{401}{225}$

Q.148. What is the difference of the largest and smallest of the given fractions ?

$$\frac{5}{11}, \frac{5}{7}, \frac{3}{8}, \frac{6}{13}$$

SSC CHSL 13/03/2023 (1st Shift)

- (a) $\frac{17}{56}$ (b) $\frac{19}{56}$ (c) $\frac{1}{7}$ (d) $\frac{23}{56}$

Q.149. What is the value of

$$\frac{(.91)^3 + (.09)^3}{[(.91)^2 - .0819 + (.09)^2]}$$

SSC CHSL 15/03/2023 (1st Shift)

- (a) 1 (b) 5 (c) 4 (d) 6

SSC CGL 2022 Tier - 2

Q.150. Find the value of the given expression.

$$\sqrt{20 - \sqrt{20 - \sqrt{20 - \sqrt{20 - \dots}}}}$$

SSC CGL Tier II (03/03/2023)

- (a) 4 (b) 6 (c) 5 (d) 2

Q.151. Simplify the following.

$$\sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + \sqrt{225}}}}}$$

$$\sqrt{16 + 19.25 \times 4^2}$$

SSC CGL Tier II (06/03/2023)

- (a) $\frac{7}{18}$ (b) $\frac{2}{9}$ (c) $\frac{5}{18}$ (d) $\frac{1}{9}$

SSC CGL 2022 Tier - 1

Q.152. Simplify $x^9 \times x^5 \times x^{-4} \times x^0 \times x^{-6}$

SSC CGL 01/12/2022 (2nd Shift)

- (a) x^4 (b) x^{-4} (c) x^{-6} (d) x^6

Q.153. What is the value of $100^2 - 99^2 + 98^2 - 97^2 + 96^2 - 95^2 + 94^2 - 93^2 + \dots + 12^2 - 11^2$?

SSC CGL 05/12/2022 (3rd Shift)

- (a) 5050 (b) 4985 (c) 4995 (d) 4950

Q.154. Simplify $\sqrt{36x^2 - 108x + 81}$.

SSC CGL 07/12/2022 (3rd Shift)

- (a) $6x - 9$ (b) $2x - 9$ (c) $5x - 9$ (d) $3x - 3$

SSC CPO 2022 Tier - 1

Q.155. The value of

$$\frac{\sqrt[3]{-2744} \times \sqrt[3]{-216}}{\sqrt[3]{\frac{64}{729}}}$$
 is

SSC CPO 09/11/2022 (Morning)

- (a) 164 (b) 152 (c) 189 (d) 156

Q.156. What is the value of 'a' in the below equation ?

$$\{(5 \times 5 \times 5 \times 5 \times 5 \times 5) \times (5 \times 5 \times 5 \times 5 \times 5) \times (5 \times 5) = (625)^a\}$$

SSC CPO 09/11/2022 (Afternoon)

- (a) 14 (b) 17 (c) 16 (d) 15

SSC CGL 2021 Tier - 2

Q.157. What is the value of $99\frac{11}{99} + 99$

$$\frac{13}{99} + 99\frac{15}{99} + \dots + 99\frac{67}{99} ?$$

SSC CGL Tier II (08/08/2022)

- (a) $\frac{94220}{33}$ (b) $\frac{95120}{33}$
(c) $\frac{97120}{33}$ (d) $\frac{96220}{33}$

Q.158. What is the simplified value of

$$\frac{(x + y + z)(xy + yz + zx) - xyz}{(x + y)(y + z)(z + x)} ?$$

SSC CGL Tier II (08/08/2022)

- (a) y (b) z (c) 1 (d) x

Q.159. What is the value of

$$\frac{\sqrt{29.16}}{\sqrt{1.1664}} + \frac{\sqrt{0.2916}}{\sqrt{116.64}} + \frac{\sqrt{0.0036}}{\sqrt{0.36}} ?$$

SSC CGL Tier II (08/08/2022)

- (a) $\frac{26}{5}$ (b) $\frac{103}{20}$ (c) $\frac{27}{5}$ (d) $\frac{101}{20}$

Q.160. If $A = \frac{\sqrt{0.0004} \times \sqrt[3]{0.000008}}{\sqrt[4]{16000} \times \sqrt[3]{125000} \times \sqrt[4]{810}}$ and $B = \frac{\sqrt[3]{0.729} \times \sqrt[4]{0.0016}}{\sqrt{0.16}}$, then what is $A \times B$?

SSC CGL Tier II (08/08/2022)

- (a) 7×10^{-7} (b) $\left(\frac{7}{4}\right) \times 10^{-8}$

- (c) $\frac{3}{5} \times 10^{-7}$ (d) $\left(\frac{7}{3}\right) \times 10^{-7}$

Q.161. If $A = 0.3\overline{12}$, $B = 0.4\overline{15}$ & $C = 0.30\overline{9}$, then what is the value of $A + B + C$?

SSC CGL Tier II (08/08/2022)

- (a) $\frac{1141}{1100}$ (b) $\frac{1097}{1100}$ (c) $\frac{1211}{1100}$ (d) $\frac{1043}{1100}$

SSC Selection Post (Phase - X)

Q.162. If $\sqrt{5} = 2.24$, then find the value of

$$\sqrt{\frac{\sqrt{5} - 1}{\sqrt{5} + 1}}$$

Graduate Level 04/08/2022 (Shift - 4)

- (a) 0.62 (b) 0.59 (c) 0.68 (d) 0.55

Q.163. Find the value of : $13^3 + (-8)^3 + (-5)^3$

Higher Secondary 01/08/2022 (Shift - 3)

- (a) 1052 (b) 1152 (c) 1562 (d) 1560

SSC MTS 2021 Tier - 1

Q.164. Select the number that will come in place of the question mark(?) in the mathematical statement.

$$(0.064)^{123} \div (0.16)^{47} \times (0.4)^{34} \times (0.4)^{29} = (0.4)^?$$

SSC MTS 08/07/2022 (Afternoon)

- (a) 350 (b) 320 (c) 338 (d) 341

Q.165. If $P = 7 \times 4 - 3$, $Q = 8 \div 4 + 2$, and $R = 14 \div 21 \times 9$, then what is the value of $P \times Q + R$?

SSC MTS 25/07/2022 (Morning)

- (a) 96 (b) 112 (c) 108 (d) 106

SSC CHSL 2021 Tier - 1

Q.166. Simplify the expression

$$(36p^2 + 49q^2)(6p + 7q)(6p - 7q)$$

SSC CHSL 24/05/2022 (Morning)

- (a) $1296p^4 + 2401q^4$ (b) $36p^4 - 49q^4$
 (c) $1296p^4 - 2401q^4$ (d) $36p^4 + 49q^4$

Q.167. Simplify :

$$3p - [3p - \overline{p + q} - \{3p - (p - \overline{q - p})\}]$$

SSC CHSL 24/05/2022 (Afternoon)

- (a) $3p - q$ (b) $p + 2q$
 (c) $3p + 2q$ (d) $2p + 2q$

Q.168. Find the value of

$$\left(\frac{9^3 + 9^4 + 9^5 + 9^6 + 9^7}{9^1 + 9^2 + 9^3 + 9^4 + 9^5}\right)^{\frac{1}{2}} = ?$$

SSC CHSL 01/06/2022 (Morning)

- (a) 9 (b) 3 (c) 729 (d) 81

Q.169. Simplify the following expression

$$3\frac{6}{7} - \{5\frac{1}{7} - (2\frac{3}{7} - 3\frac{5}{7} - 4)\}$$

SSC CHSL 01/06/2022 (Afternoon)

- (a) $\frac{34}{7}$ (b) $\frac{10}{7}$ (c) $\frac{20}{7}$ (d) $\frac{44}{7}$

Q.170. If a, b, c are positive real numbers, what is the value of

$$\sqrt{a^{-1}b} \times \sqrt{b^{-1}c} \times \sqrt{c^{-1}a} ?$$

SSC CHSL 03/06/2022 (Evening)

- (a) 2 (b) 3 (c) 1 (d) 4

Q.171. The value of

$$\frac{(a-b)^3 + (b-c)^3 + (c-a)^3}{(a^2 - b^2)^3 + (b^2 - c^2)^3 + (c^2 - a^2)^3}$$

is equal to

SSC CHSL 06/06/2022 (Morning)

- (a) $ab + bc + ca$ (b) 0
 (c) 1 (d) $\frac{1}{(a+b)(b+c)(c+a)}$

Q.172. Find the value of

$$\sqrt{2} + \sqrt{3} + \sqrt{2} - \sqrt{3}.$$

SSC CHSL 06/06/2022 (Evening)

- (a) $\sqrt{6}$ (b) $2\sqrt{3}$ (c) $2\sqrt{2}$ (d) 6

Q.173. Simplify :

$$\frac{(r-s)^3 + (s-t)^3 + (t-r)^3}{6(r-s)(s-t)(t-r)}$$

SSC CHSL 09/06/2022 (Afternoon)

- (a) $\frac{1}{6}$ (b) $\frac{1}{2}$ (c) $\frac{1}{3}$ (d) $\frac{1}{4}$

Q.174. Simplify: $(a^{-1} + b^{-1}) \div (a^{-3} + b^{-3})$

SSC CHSL 09/06/2022 (Evening)

- (a) $\frac{ab}{a^2 - ab + b^2}$ (b) $\frac{a^3 b^3}{a^2 - ab + b^2}$
 (c) $\frac{a^2 b^2}{a^2 + ab + b^2}$ (d) $\frac{a^2 b^2}{a^2 - ab + b^2}$

SSC CGL 2021 Tier - 1

Q.175. Simplify the following expressions:

$$\frac{(a^2 - 4b^2)^3 + 64(b^2 - 4c^2)^3 + (16c^2 - a^2)^3}{(a - 2b)^3 + (2b - 4c)^3 + (4c - a)^3}$$

SSC CGL 11/04/2022 (Afternoon)

- (a) $-(a - 2b)(b + 2c)(4c + a)$
 (b) $2(a + 2b)(b + 2c)(4c + a)$
 (c) $(a + 2b)(b + 2c)(4c + a)$
 (d) $4(a + 2b)(b + 2c)(4c + a)$

Q.176. If $5\sqrt{3} + \sqrt{75} = 17.32$, then the value of $14\sqrt{3} + \sqrt{108}$ is :

SSC CGL 20/04/2022 (Evening)

- (a) 32.46 (b) 35.64 (c) 34.64 (d) 33.86

SSC CGL 2020 Tier - 2

Q.177. If $847 \times 385 \times 675 \times 3025 = 3^a \times 5^b \times 7^c \times 11^d$, then the value of $ab - cd$ is:

SSC CGL Tier II (29/01/2022)

- (a) 5 (b) 1 (c) 4 (d) 7

Q.178. The value of

$$\frac{1}{4 - \sqrt{15}} - \frac{1}{\sqrt{15} - \sqrt{14}} + \frac{1}{\sqrt{14} - \sqrt{13}} - \frac{1}{\sqrt{13} - \sqrt{12}} + \frac{1}{\sqrt{12} - \sqrt{11}} - \frac{1}{\sqrt{11} - \sqrt{10}} + \frac{1}{\sqrt{10} - 3} - \frac{1}{3 - \sqrt{8}}$$

is:

SSC CGL Tier II (29/01/2022)

- (a) $4 - 2\sqrt{2}$ (b) $2 + 2\sqrt{2}$
 (c) $4 + 2\sqrt{2}$ (d) $2 - 2\sqrt{2}$

Q.179. If $\frac{\sqrt{38 - 5\sqrt{3}}}{\sqrt{26 + 7\sqrt{3}}} = \frac{a + b\sqrt{3}}{23}$, $b > 0$,

then the value of $(b - a)$ is :

SSC CGL Tier II (03/02/2022)

- (a) 7 (b) 18 (c) 29 (d) 11

SSC MTS 2020 Tier - 1

Q.180. What is the simplified value of

$$\{4 - \frac{2}{1 + \frac{2}{1 - \frac{1}{2 + \frac{3}{4}}}}\} \div 1\frac{5}{12} \text{ of } \frac{72}{145} - (4$$

$$+ 3 \div 0.5 - 1)$$

SSC MTS 20/10/2021 (Afternoon)

- (a) -4 (b) 1 (c) 3 (d) -2

Q.181. The value of $4\frac{2}{5} \div \{(1\frac{1}{2} - 3\frac{1}{5})$

$$\div 3\frac{2}{5} + (2\frac{1}{5} \div 1\frac{1}{2} + 4\frac{2}{5}) + \frac{1}{2}\} = ?$$

SSC MTS 22/10/2021 (Evening)

- (a) $\frac{4}{7}$ (b) $\frac{3}{7}$ (c) $\frac{3}{5}$ (d) $\frac{3}{4}$

SSC CGL 2020 Tier - 1

Q.182. The value of $3\frac{5}{6} + [3\frac{2}{3} + \{ \frac{15}{4} (5$

$$\frac{4}{5} \div 14\frac{1}{2})]$$

is equal to:

SSC CGL 16/08/2021 (Afternoon)

- (a) 9 (b) 6 (c) 7 (d) 8

Q.183. Simplify the following expression:

$$(\frac{3}{4} - \frac{1}{4} \div \frac{1}{4} \text{ of } \frac{2}{5}) \div (\frac{3}{4} \div \frac{2}{3} \text{ of } \frac{3}{5})$$

SSC CGL 24/08/2021 (Evening)

- (a) $\frac{14}{75}$ (b) $-\frac{70}{27}$ (c) $-\frac{14}{15}$ (d) $\frac{32}{75}$

SSC CHSL 2020 Tier - 1

Q.184. Simplify the following expression.

$$(2x - 3y)^3 - 18xy(2x - 3y)$$

SSC CHSL 12/04/2021 (Morning)

- (a) $8x^3 - 27y^3 - 36x^2y - 54xy^2$
 (b) $8x^3 - 27y^3 - 72x^2y + 108xy^2$
 (c) $8x^3 - 27y^3$
 (d) $8x^3 + 108xy^2 - 72x^2y$

Q.185. The value of

$$364 - [142 - \{75 + (38 - \frac{5}{4} + \frac{14}{8})\}]$$

is:

SSC CHSL 11/08/2021 (Evening)

- (a) 332 (b) 330 (c) 331 (d) 334

Answer Key :-

1.(b)	2.(a)	3.(a)	4.(b)
5.(a)	6.(a)	7.(c)	8.(b)
9.(d)	10.(a)	11.(b)	12.(d)
13.(c)	14.(c)	15.(d)	16.(d)
17.(d)	18.(a)	19.(c)	20.(a)
21.(c)	22.(a)	23.(b)	24.(d)
25.(a)	26.(d)	27.(d)	28.(d)
29.(a)	30.(b)	31.(c)	32.(b)
33.(b)	34.(b)	35.(b)	36.(a)
37.(d)	38.(b)	39.(b)	40.(d)
41.(a)	42.(d)	43.(d)	44.(a)
45.(b)	46.(d)	47.(b)	48.(c)
49.(a)	50.(d)	51.(d)	52.(a)
53.(b)	54.(b)	55.(d)	56.(b)
57.(c)	58.(c)	59.(a)	60.(d)
61.(d)	62.(b)	63.(b)	64.(a)
65.(c)	66.(c)	67.(c)	68.(d)
69.(b)	70.(b)	71.(c)	72.(d)
73.(a)	74.(d)	75.(b)	76.(c)
77.(a)	78.(b)	79.(c)	80.(b)
81.(a)	82.(c)	83.(a)	84.(d)
85.(b)	86.(a)	87.(d)	88.(c)
89.(a)	90.(c)	91.(b)	92.(c)
93.(b)	94.(c)	95.(b)	96.(d)
97.(b)	98.(a)	99.(c)	100.(c)
101.(a)	102.(b)	103.(a)	104.(d)
105.(d)	106.(c)	107.(b)	108.(c)

109.(d)	110.(c)	111.(a)	112.(d)
113.(d)	114.(c)	115.(d)	116.(b)
117.(c)	118.(b)	119.(a)	120.(c)
121.(d)	122.(c)	123.(b)	124.(d)
125.(c)	126.(d)	127.(b)	128.(d)
129.(c)	130.(b)	131.(a)	132.(a)
133.(d)	134.(c)	135.(c)	136.(b)
137.(c)	138.(d)	139.(b)	140.(b)
141.(b)	142.(d)	143.(b)	144.(c)
145.(a)	146.(d)	147.(a)	148.(b)
149.(a)	150.(a)	151.(b)	152.(a)
153.(c)	154.(a)	155.(c)	156.(b)
157.(b)	158.(c)	159.(b)	160.(c)
161.(a)	162.(a)	163.(d)	164.(c)
165.(d)	166.(c)	167.(d)	168.(a)
169.(b)	170.(c)	171.(d)	172.(a)
173.(b)	174.(d)	175.(b)	176.(c)
177.(a)	178.(a)	179.(c)	180.(a)
181.(d)	182.(a)	183.(c)	184.(b)
185.(a)			

Solutions :-

$$\begin{aligned}\text{Sol.1.(b)} & \frac{(25)^4 + 1250 + 1 - (25)^2}{(25)^2 + 25 + 1} \\ & = \frac{(25)^4 - (25)^2 + 1250 + 1}{(25)^2 + 25 + 1} \\ & = \frac{[(25)^2]^2 + 2 \times 25^2 \times 1 + (1)^2 - (25)^2}{(25)^2 + 25 + 1} \\ & = \frac{[(25)^2 + 1]^2 - (25)^2}{(25)^2 + 25 + 1} \\ & = \frac{[(25^2 + 1 + 25)(25^2 + 1 - 25)]}{(25)^2 + 25 + 1} \\ & = (25^2 + 1 - 25) = 625 - 24 = 601\end{aligned}$$

$$\begin{aligned}\text{Sol.2.(a)} & \frac{25a^2 - 10ab - 48b^2}{5a + 6b} \times (5a + 8b) \\ & \frac{25a^2 - 40ab + 30ab - 48b^2}{5a + 6b} \times (5a + 8b) \\ & \frac{5a(5a - 8b) + 6b(5b - 8b)}{5a + 6b} \times (5a + 8b) \\ & \frac{(5a - 8b)(5a + 6b)}{5a + 6b} \times (5a + 8b) \\ & (5a - 8b) \times (5a + 8b) \\ & (5a)^2 - (8b)^2 = 25a^2 - 64b^2\end{aligned}$$

$$\begin{aligned}\text{Sol.3.(a)} & \frac{(x^2 - 9) \times (9x^2 - 1)}{(x - \frac{1}{3}) \times (1 - \frac{x}{3})} \times \frac{9}{(x + \frac{1}{3})(1 + \frac{x}{3})} \\ & = \frac{(x + 3)(x - 3)(3x + 1)(3x - 1)}{\frac{(3x - 1)(3 - x)}{9}} \times\end{aligned}$$

$$\frac{9}{\frac{(3x + 1)(3 + x)}{9}} = -729$$

Sol.4.(b)

$$\begin{aligned}A & = \frac{58^2 - 25^2}{46^2 - 37^2} = \frac{(58 - 25)(58 + 25)}{(46 - 37)(46 + 37)} \\ & = \frac{33 \times 83}{9 \times 83} = \frac{11}{3}\end{aligned}$$

$$B = \frac{26^2 - 15^2}{56^2 - 15^2} = \frac{(26 - 15)(26 + 15)}{(56 - 15)(56 + 15)}$$

$$= \frac{11 \times 41}{41 \times 71} = \frac{11}{71}$$

$$\text{Then, } \frac{1}{B} - \frac{20}{A} = \frac{71}{11} - \frac{20 \times 3}{11} = \frac{11}{11} = 1$$

Sol.5.(a)

$$\Rightarrow [(2\frac{7}{9})^2]^{\frac{3}{5}} = [(\frac{5}{3})^2]^{\frac{5}{2}}]^{\frac{3}{5}} = (\frac{5}{3})^3$$

$$\Rightarrow [(1\frac{2}{3})^5]^{\frac{3}{5}} = (\frac{5}{3})^3$$

∴ both the expressions are equal

Sol.6.(a)

$$\begin{aligned}& \frac{24\sqrt{45} - 16\sqrt{20} + \sqrt{245} - 47\sqrt{5}}{3\frac{1}{5} + 1\frac{2}{5} - 2\frac{1}{6} - 2\frac{1}{5} - 5\frac{2}{5} + 3\frac{1}{6}} \\ & = \frac{72\sqrt{5} - 32\sqrt{5} + 7\sqrt{5} - 47\sqrt{5}}{\frac{16}{5} + \frac{7}{5} - \frac{13}{6} - \frac{11}{5} - \frac{27}{5} + \frac{19}{6}} \\ & = \frac{79\sqrt{5} - 79\sqrt{5}}{\frac{16}{5} + \frac{7}{5} - \frac{13}{6} - \frac{11}{5} - \frac{27}{5} + \frac{19}{6}} = 0\end{aligned}$$

Sol.7.(c)

$$\begin{aligned}& \frac{(3.321)^3 + (2.681)^3 + (1.245)^3 - 3 \times 3.321 \times 2.681 \times 1.245}{(3.321)^2 + (2.681)^2 + (1.245)^2 - (3.321 \times 2.681) - (2.681 \times 1.245) - (1.245 \times 3.321)} \\ & \frac{a^3 + b^3 + c^3 - 3abc}{a^2 + b^2 + c^2 - ab - bc - ca} = (a + b + c)\end{aligned}$$

Here, $3.321 + 2.681 + 1.245 = 7.247$

$$\begin{aligned}\text{Sol.8.(b)} & 0.01 + 0.68 \times 0.68 + 0.22 \times 0.22 \\ & + 0.44 + 0.136 + 0.44 \times 0.68 \\ & = (0.68 + 0.22)^2 + 0.01 + 0.044 + 0.136 \\ & 0.81 + 0.19 = 1\end{aligned}$$

$$\text{Sol.9.(d)} (a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

$$\frac{0.09 \times 0.09 + 0.04 \times 0.04 + 0.16 \times 0.16 + 2 \times 0.09 \times 0.04 + 2 \times 0.04 \times 0.16 + 2 \times 0.09 \times 0.16}{0.3 \times 0.3 + 0.2 \times 0.2 + 0.4 \times 0.4}$$

$$\begin{aligned}& \frac{(0.09)^2 + (0.04)^2 + (0.16)^2 + 2 \times 0.09 \times 0.04 + 2 \times 0.04 \times 0.16 + 2 \times 0.09 \times 0.16}{0.09 + 0.04 + 0.16} \\ & = \frac{(0.09 + 0.04 + 0.16)^2}{0.09 + 0.04 + 0.16} \\ & = 0.09 + 0.04 + 0.16 = 0.29\end{aligned}$$

$$\begin{aligned}\text{Sol.10.(a)} & \frac{4^{a+4} - 5 \times 4^{a+2}}{15 \times 4^a - 2^2 \times 4^a} \\ & = \frac{4^a(4^4 - 5 \times 4^2)}{4^a(15 - 2^2)} = \frac{(256 - 80)}{11} = 16\end{aligned}$$

Sol.11.(b)

$$(2^2 + 1)(2^4 + 1)(2^8 + 1)(2^{16} + 1) \dots (2^{128} + 1)$$

Multiply and divide by $(2^2 - 1)$ in the given equation, we get

$$\frac{(2^2 - 1)(2^2 + 1)(2^4 + 1)(2^8 + 1)(2^{16} + 1) \dots (2^{128} + 1)}{(2^2 - 1)}$$

$$\begin{aligned}& = \frac{(2^4 - 1)(2^4 + 1)(2^8 + 1)(2^{16} + 1) \dots (2^{128} + 1)}{3} \\ & = \frac{(2^8 - 1)(2^8 + 1)(2^{16} + 1) \dots (2^{128} + 1)}{3} \\ & = \frac{(2^{256} - 1)}{3}\end{aligned}$$

$$\text{Sol.12.(d)} \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \dots + \frac{1}{110} < \frac{5}{6}$$

$$= (1 - \frac{1}{2}) + (\frac{1}{2} - \frac{1}{3}) + (\frac{1}{3} - \frac{1}{4}) + \dots$$

$$(\frac{1}{10} - \frac{1}{11}) < \frac{5}{6} = 1 - \frac{1}{11} = \frac{10}{11} > \frac{5}{6}$$

Hence, the first statement is wrong.

$$\text{now, } \frac{1}{3} + \frac{1}{15} + \frac{1}{35} + \dots + \frac{1}{143} > \frac{7}{13}$$

$$= \frac{1}{2} [(1 - \frac{1}{3}) + (\frac{1}{3} - \frac{1}{5}) + (\frac{1}{5} - \frac{1}{7})$$

$$+ \dots + (\frac{1}{11} - \frac{1}{13})] > \frac{7}{13} = \frac{1}{2} [(1 - \frac{1}{13})$$

$$= \frac{6}{13} < \frac{7}{13}$$

Hence, the second statement is also wrong.

Sol.13.(c)

$$[\frac{5}{8} - \{\frac{3}{8} - (\frac{5}{8} - \frac{3}{8})\}] \text{ of } 8.8 - 1.2$$

$$\frac{4\frac{1}{6} \div 2.5 \times 2 \div \frac{1}{6} \text{ of } 60 + (\frac{3}{4} - \frac{3}{8})}{[\frac{5}{8} - \{\frac{3}{8} - \frac{1}{4}\}] \text{ of } 8.8 - 1.2}$$

$$= \frac{\frac{25}{6} \div 2.5 \times 2 \div 10 + \frac{3}{8}}{[\frac{5}{8} - \frac{1}{8}] \text{ of } 8.8 - 1.2}$$

$$= \frac{\frac{25}{6} \div \frac{5}{2} \times 2 \div 10 + \frac{3}{8}}{[\frac{5}{8} - \frac{1}{8}] \text{ of } 8.8 - 1.2}$$

$$= \frac{\frac{1}{2} \text{ of } 8.8 - 1.2}{\frac{5}{3} \times \frac{1}{5} + \frac{3}{8}} = \frac{4.4 - 1.2}{\frac{1}{3} + \frac{3}{8}}$$

$$= \frac{3.2}{\frac{17}{24}} = 4\frac{44}{85}$$

$$= \frac{3.2}{\frac{17}{24}} = 4\frac{44}{85}$$

$$= \frac{3.2}{\frac{17}{24}} = 4\frac{44}{85}$$

$$= \frac{3.2}{\frac{17}{24}} = 4\frac{44}{85}$$

Sol.14.(c)

$$\frac{24 \div \frac{3}{8} \text{ of } (8 + 2 \times \sqrt{7-3}) + [\frac{2}{11} \div \frac{4}{55} - \{\frac{5}{8} + \frac{6}{16}\}]}{32 \div 15 - 7 + 75 \div (6 + 15 \div 3 + 4)}$$

$$\frac{24 \div \frac{3}{8} \text{ of } (8 + 2 \times 4) + [\frac{5}{2} - 1]}{32 \div 8 + 75 \div (6 + 5 + 4)}$$

$$\frac{24 \div \frac{3}{8} \text{ of } 16 + 1.5}{4 + 75 \div 15} = \frac{4 + 1.5}{9}$$

$$= \frac{5.5}{9} = \frac{11}{18}$$

$$= \frac{5.5}{9} = \frac{11}{18}$$

Sol.15.(d)

$$\begin{aligned}& \frac{(6.9)^3 + (2.7)^3 + (5.8)^3 - 3 \times 6.9 \times 2.7 \times 5.8}{(6.9)^2 + (2.7)^2 + (5.8)^2 - (6.9)(2.7) - (2.7)(5.8) - (5.8)(6.9)} \\ & \frac{(6.9 + 2.7 + 5.8)((6.9)^2 + (2.7)^2 + (5.8)^2 - (6.9 \times 2.7) - (2.7 \times 5.8) - (5.8 \times 6.9))}{(6.9)^2 + (2.7)^2 + (5.8)^2 - (6.9)(2.7) - (2.7)(5.8) - (5.8)(6.9)}\end{aligned}$$

$$= 6.9 + 2.7 + 5.8 = 15.4$$

Sol.16.(d)

$$9\frac{1}{5} \div \{(16\frac{1}{5} \div 12\frac{2}{3} - 9\frac{2}{3} + 17\frac{1}{2}$$

$$\text{of } 3\frac{1}{21}\}$$

$$= \frac{46}{5} \div \left\{ \frac{81}{5} \div \left(\frac{38}{3} - \frac{29}{3} \right) + \frac{35}{2} \right\}$$

$$\text{of } \frac{64}{21}$$

$$= \frac{46}{5} \div \left\{ \frac{81}{5} \div 3 + \frac{160}{3} \right\}$$

$$= \frac{46}{5} \div \left\{ \frac{27}{5} + \frac{160}{3} \right\}$$

$$= \frac{46}{5} \div \frac{881}{15} \Rightarrow \frac{46}{5} \times \frac{15}{881} \Rightarrow \frac{138}{881}$$

Sol.17.(d)

$$\frac{1}{4} + \frac{[(20.35)^2 - (8.35)^2] \times 0.0175}{(1.05)^2 + (1.05)(27.65)}$$

$$= \frac{1}{4} + \frac{[(20.35+8.35)(20.35-8.35)] \times 0.0175}{(1.05)\{(1.05) + (27.65)\}}$$

$$= \frac{1}{4} + \frac{[28.7 \times 12] \times 0.0175}{(1.05)(28.7)}$$

$$= \frac{1}{4} + \frac{12 \times 175}{105 \times 100} = \frac{1}{4} + \frac{12 \times 5}{3 \times 100}$$

$$= \frac{1}{4} + \frac{1}{5} = \frac{9}{20}$$

Sol.18.(a)

$$\frac{119 - [48 \div 6 - 7 \{5 \times 12 \div 3 - (15 - 3 - 8)\}]}{24 \div 3 - 9 \text{ of } 3 + (52 - 8 - 4) \div 6}$$

$$= \frac{119 - [8 - 7 \{5 \times 4 - (15 + 5)\}]}{8 - 27 + (52 - 4) \div 6}$$

$$= \frac{119 - [8 - 7 \{20 - (20)\}]}{8 - 27 + (48) \div 6}$$

$$= \frac{119 - [8 - 7 \{0\}]}{8 - 27 + 8}$$

$$= \frac{119 - [8]}{-11} = -\frac{111}{11}$$

Sol.19.(c) $\frac{1}{4 - \sqrt{8}}$ (on rationalizing)

$$= \frac{1}{4 - \sqrt{8}} \times \frac{4 + \sqrt{8}}{4 + \sqrt{8}} = \frac{4 + \sqrt{8}}{8} \text{ so,}$$

$$= \frac{4 + \sqrt{8}}{8} + \frac{3 + 2\sqrt{2}}{3 - 2\sqrt{2}} - \frac{3 - 2\sqrt{2}}{3 + 2\sqrt{2}}$$

$$= \frac{4 + \sqrt{8}}{8} + \frac{(3 + 2\sqrt{2})^2 - (3 - 2\sqrt{2})^2}{(3)^2 - (2\sqrt{2})^2}$$

$$= \frac{4 + \sqrt{8}}{8} + \frac{24\sqrt{2}}{9 - 8} \Rightarrow \frac{1}{2} + \frac{\sqrt{2}}{4} + 24\sqrt{2}$$

$$\text{Since, } \frac{1}{2} + \frac{97}{4}\sqrt{2} = a + b\sqrt{2}$$

$$\text{On comparing } a = \frac{1}{2}, b = \frac{97}{4}$$

$$3a + 4b = 3 \times \frac{1}{2} + 4 \times \frac{97}{4} = 98\frac{1}{2}$$

Sol.20.(a) After checking option one by one option (a) satisfies.

$$4 \times 30 - 15 \div 5 + 12 = 21$$

$$\Rightarrow 4 \times 30 \div 5 - 15 + 12 = 21$$

$$\Rightarrow 4 \times 6 - 15 + 12 = 21$$

$$\Rightarrow 24 - 15 + 12 = 21 \Rightarrow 9 + 12 = 21$$

$$\Rightarrow 21 = 21$$

Sol.21.(c) $\frac{4}{1 + \sqrt{2} + \sqrt{3}}$

$$= a + b\sqrt{2} + c\sqrt{3} - d\sqrt{6}$$

$$= \frac{4}{1 + \sqrt{2} + \sqrt{3}} \times \frac{\sqrt{3} + 1 - \sqrt{2}}{\sqrt{3} + 1 - \sqrt{2}}$$

$$= (a + b)(a - b) = a^2 - b^2$$

$$\frac{4\sqrt{3} + 4 - 4\sqrt{2}}{(\sqrt{3} + 1)^2 - (\sqrt{2})^2} = \frac{4\sqrt{3} + 4 - 4\sqrt{2}}{2\sqrt{3} + 2}$$

$$= \frac{2\sqrt{3} + 2 - 2\sqrt{2}}{\sqrt{3} + 1} \times \frac{\sqrt{3} - 1}{\sqrt{3} - 1}$$

$$= \frac{(2\sqrt{3} + 2 - 2\sqrt{2})(\sqrt{3} - 1)}{3 - 1}$$

$$= \frac{(2\sqrt{3} + 2 - 2\sqrt{2})(\sqrt{3} - 1)}{2}$$

$$= (\sqrt{3} + 1 - \sqrt{2})(\sqrt{3} - 1)$$

$$= 3 + \sqrt{3} - \sqrt{6} - \sqrt{3} - 1 + \sqrt{2}$$

$$= 2 + \sqrt{2} - \sqrt{6} = a + b\sqrt{2} + c\sqrt{3} - d\sqrt{6}$$

$$a = 2; b = 1; c = 0; d = 1$$

$$a + b + c + d = 2 + 1 + 0 + 1 = 4$$

Sol.22.(a)

$$\left[\left(\frac{2}{3} \right)^3 (2x + 3) \right]^{\frac{-3}{4}} = \left[\left(\frac{2}{3} \right)^{\frac{2}{3}} (3x + 7) \right]^{\frac{-6}{5}}$$

$$\Rightarrow \left[\left(\frac{2}{3} \right)^3 (2x + 3) \right]^{\frac{3}{4}} = \left[\left(\frac{2}{3} \right)^{\frac{2}{3}} (3x + 7) \right]^{\frac{6}{5}}$$

$$\Rightarrow \left[\left(\frac{2}{3} \right)^3 \right]^{\frac{3(2x+3)}{4}} = \left[\left(\frac{2}{3} \right)^{\frac{2}{3}} \right]^{\frac{6(3x+7)}{5}}$$

$$\Rightarrow \left[\left(\frac{2}{3} \right)^3 \right]^{\frac{6x+9}{4}} = \left[\left(\frac{2}{3} \right)^{\frac{2}{3}} \right]^{\frac{18x+42}{5}}$$

$$\Rightarrow \left(\frac{2}{3} \right)^{\frac{3(6x+9)}{4}} = \left(\frac{2}{3} \right)^{\frac{2(18x+42)}{5}}$$

$$= \left[\left(\frac{2}{3} \right)^{\frac{2(18x+42)}{3 \times 5}} \right] = \left[\left(\frac{2}{3} \right)^{\frac{18x+27}{5}} \right]$$

$$= \frac{36x + 84}{15} = \frac{18x + 27}{5}$$

$$\Rightarrow 4(36x + 84) = 15(18x + 27)$$

$$\Rightarrow 144x + 336 = 270x - 405$$

$$\Rightarrow 126x = 336 - 405 \Rightarrow 126x = -69$$

$$\Rightarrow x = -\frac{69}{126} = -\frac{23}{42}$$

Now put the value of x in given equation

$$\sqrt{2 - 42x} = \sqrt{2 - 42\left(-\frac{23}{42}\right)}$$

$$= \sqrt{2 + 23} = \sqrt{25} = 5$$

Sol.23.(b) $\frac{42 - 12 \times 3 + 8 \div 2 + 15}{8 \times 2 - 4 + 9 \div 3}$

After applying changes as per question we get

$$\frac{42 + 12 \div 3 - 8 \times 2 - 15}{8 \div 2 + 4 - 9 \times 3} = -\frac{15}{19}$$

$$\text{Sol.24.(d)} \frac{(4.6)^4 + (5.4)^4 + (24.84)^2}{(4.6)^2 + (5.4)^2 + 24.84}$$

$$\frac{\{(4.6)^2 + (5.4)^2\}^2 - 2\{(4.6)^2(5.4)^2\} + (24.84)^2}{(4.6)^2 + (5.4)^2 + 24.84}$$

$$= \frac{\{(4.6)^2 + (5.4)^2\}^2 - (24.84)^2}{(4.6)^2 + (5.4)^2 + 24.84}$$

$$= \frac{[(4.6)^2 + (5.4)^2] - (24.84)}{(4.6)^2 + (5.4)^2 + 24.84}$$

$$= (4.6)^2 + (5.4)^2 - 24.84$$

$$= (4.6 + 5.4)^2 - 2(4.6)(5.4) - 24.84$$

$$= 100 - 74.52 = 25.48$$

$$\text{Sol.25.(a)} \frac{(0.545)(0.081)(0.51)(5.2)}{(0.324)^3 + (0.221)^3 - (0.545)^3}$$

$$= \frac{(0.545)(0.081) \times 4 \times 3 \times (0.17)(1.3)}{(0.324)^3 + (0.221)^3 - (0.545)^3}$$

$$= \frac{3 \times (0.545)(0.324)(0.221)}{(0.324)^3 + (0.221)^3 - (0.545)^3}$$

Now, since

$$(0.324 + 0.221 - 0.545 = 0)$$

$$\Rightarrow (0.324)^3 + (0.221)^3 - (0.545)^3 =$$

$$- \{3 \times (0.545)(0.324)(0.221)\}$$

$$\dots (\text{Algebraic property})$$

$$\text{So, } \frac{3 \times (0.545)(0.324)(0.221)}{(0.324)^3 + (0.221)^3 - (0.545)^3}$$

$$= \frac{3 \times (0.545)(0.324)(0.221)}{-\{3 \times (0.545)(0.324)(0.221)\}} = -1$$

Sol.26.(d) $22.\bar{4} + 11.5\overline{67} - 33.\bar{59}$

$$= 22.444444\dots\infty + 11.5676767\dots\infty -$$

$$33.599999\dots\infty$$

$$= 0.412121212\dots\infty = 0.4\bar{12}$$

Sol.27.(d)

$$\frac{(0.7)^2 \div 0.14 + (0.6)^2 \div 0.18 + (0.5)^2 \div 0.05}{4(2.5 \text{ of } 4 - 13 \times 0.25 \times 3)}$$

$$= \frac{0.49 \div 0.14 + 0.36 \div 0.18 + 0.25 \div 0.05}{4(10 - 9.75)}$$

$$= \frac{3.5 + 2 + 5}{4 \times 0.25} = 10.5 = \frac{21}{2}$$

Sol.28.(d)

$$x = \frac{1}{12.13} + \frac{1}{13.14} + \frac{1}{14.15} \dots + \frac{1}{23.24}$$

$$= \frac{1}{12} - \frac{1}{13} + \frac{1}{13} - \frac{1}{14} + \dots + \frac{1}{23} - \frac{1}{24}$$

$$= \frac{1}{12} - \frac{1}{24} = \frac{1}{24}$$

$$y = \frac{1}{36.37} + \frac{1}{37.38} + \frac{1}{38.39} \dots + \frac{1}{71.72}$$

$$= \frac{1}{36} - \frac{1}{37} + \frac{1}{37} - \frac{1}{38} + \dots + \frac{1}{71} - \frac{1}{72}$$

$$= \frac{1}{36} - \frac{1}{72} = \frac{1}{72}$$

$$\text{Therefore, } \frac{x}{y} = \frac{\frac{1}{24}}{\frac{1}{72}} = 3$$

Alternate Method :-

$$x = \frac{1}{12.13} + \frac{1}{13.14} + \frac{1}{14.15} \dots + \frac{1}{23.24}$$

$$= \frac{1}{13 - 12} \left(\frac{1}{12} - \frac{1}{24} \right) = \frac{1}{24}$$

$$y = \frac{1}{36.37} + \frac{1}{37.38} + \frac{1}{38.39} \dots + \frac{1}{71.72}$$

$$= \frac{1}{37 - 36} \left(\frac{1}{36} - \frac{1}{72} \right) = \frac{1}{72}$$

$$\text{Therefore, } \frac{x}{y} = \frac{\frac{1}{24}}{\frac{1}{72}} = 3$$

$$\text{Sol.29.(a)} \frac{0.01404}{24^2 + 6^2 - 144}$$

$$= \frac{0.01404}{576 + 36 - 144} = \frac{0.01404}{612 - 144}$$

$$= \frac{0.01404}{468} = 3 \times 10^{-5}$$

$$\text{Sol.30.(b)} \frac{1.0025 + 6.25 \times 10^{-6}}{0.0025 + 0.95} \\ = \left(\frac{1.00250625}{0.9525} \right) = 1.0525$$

Sol.31.(c)

$$5 + 24 \div 8 \div 8 \text{ of } 8 - 8 \div 8 \times 4 \text{ of } 4 \\ = 5 + 24 \div 8 \div 64 - 1 \times 16 = 5 + 3 - 16 = -8$$

Sol.32.(b)

$$(2z - 5y)^2 + (5z + 2y)^2 - 25z^2 \\ = 4z^2 + 25y^2 - 20zy + 25z^2 + 4y^2 + 20zy - 25z^2 = 29y^2 + 4z^2$$

Sol.33.(b) According to question, HCF of (221 and 488) = 1

Since the HCF is 1, the fraction $\frac{221}{488}$ is already in its lowest terms.

So, the lowest term is $\frac{221}{488}$

$$\text{Sol.34.(b)} \frac{(x^2 - 9)}{x + 3} = \frac{(x^2 - 3^2)}{x + 3} \\ = \frac{(x - 3)(x + 3)}{(x + 3)} = (x - 3)$$

Sol.35.(b)

$$\frac{(8.3)^3 + (9.2)^3 + (6.1)^3 - 3 \times 8.3 \times 9.2 \times 6.1}{(8.3)^2 + (9.2)^2 + (6.1)^2 - 8.3 \times 9.2 - 9.2 \times 6.1 - 6.1 \times 8.3} \\ \text{Identity used:- } (a)^3 + (b)^3 + (c)^3 - 3abc = (a + b + c)[(a)^2 + (b)^2 + (c)^2 - ab - bc - ac] \\ \frac{(8.3)^3 + (9.2)^3 + (6.1)^3 - 3 \times 8.3 \times 9.2 \times 6.1}{(8.3 + 9.2 + 6.1)} \\ = 8.3 + 9.2 + 6.1 = 23.6$$

$$\text{Sol.36.(a)} \frac{0.58 \times 0.58 \times 0.58 - 0.001}{0.58 \times 0.58 + 0.058 + 0.01} \\ = \frac{(0.58)^3 - (0.1)^3}{[(0.58)^2 + 0.1 \times 0.58 + (0.1)^2]}$$

Using identity :

$$(a)^3 - (b)^3 = (a - b)(a^2 + b^2 + ab) \\ = \frac{(0.58 - 0.1)[(0.58)^2 + (0.1)^2 + 0.1 \times 0.58]}{[(0.58)^2 + 0.1 \times 0.58 + (0.1)^2]} \\ = 0.58 - 0.1 = 0.48$$

Sol.37.(d)

$$45 \div 3 + 4 \times A - 48 \div 24 + 4 = 1 \\ 15 + 4A - 2 + 4 = 1 \Rightarrow 17 + 4A = 1 \\ 4A = 1 - 17 \Rightarrow 4A = -16 \Rightarrow A = -4$$

Sol.38.(b) Formula used :

$$a^3 - b^3 = (a - b)(a^2 + b^2 + ab) \\ \frac{(4.2)^3 - (1.2)^3}{(4.2)^2 + 5.04 + (1.2)^2} \\ = \frac{(4.2 - 1.2)[(4.2)^2 + 5.04 + (1.2)^2]}{(4.2)^2 + 5.04 + (1.2)^2} = 3$$

Sol.39.(b)**Formula :** $a^3 - b^3 = (a - b)(a^2 + b^2 + ab)$

Now, we have ;

$$(1.55^2 + 0.55^2 + 1.55 \times 0.55)$$

On multiplying and dividing the above

$$\text{expression by } (1.55 - 0.55), \text{ We get ;} \\ \frac{(1.55 - 0.55)(1.55^2 + 0.55^2 + 1.55 \times 0.55)}{(1.55 - 0.55)} \\ = \frac{1.55^3 - 0.55^3}{1} = 1.55^3 - 0.55^3$$

Sol.40.(d)

$$[(2.5 + 1.5)^2 \div 4 \text{ of } \frac{1}{2} \times 8 - 1] \\ [(4)^2 \div 2 \times 8 - 1] = 8 \times 8 - 1 \\ = 64 - 1 = 63$$

$$\text{Sol.41.(a)} \frac{(x-y)^3 + (y-z)^3 + (z-x)^3}{(x-y)(y-z)(z-x)}$$

We know that, when $a + b + c = 0$ then $(a)^3 + (b)^3 + (c)^3 = 3abc$.

Now,

$$x - y + y - z + z - x = 0$$

$$\text{So, } \frac{3(x-y)(y-z)(z-x)}{(x-y)(y-z)(z-x)} = 3$$

Sol.42.(d) $(g - h)^2 - (g + h)^2 + 4gh$

$$\text{Identity used : } a^2 - b^2 = (a + b)(a - b) \\ = (g - h - g - h)(g - h + g + h) + 4gh \\ = (-2h)(2g) + 4gh = -4gh + 4gh = 0$$

Sol.43.(d)

$$\frac{1.5 \times 1.5 + 2.5 \times 2.5 + 3.5 \times 3.5 + 2 \times 1.5 \times 2.5 + 2 \times 2.5 \times 3.5 + 2 \times 1.5 \times 3.5}{1.5 + 2.5 + 3.5}$$

We know that $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$

$$\frac{(1.5 + 2.5 + 3.5)^2}{1.5 + 2.5 + 3.5} = 1.5 + 2.5 + 3.5 = 7.5$$

Sol.44.(a)

$$[(5\frac{1}{2} - 3\frac{1}{3}) \text{ of } 1\frac{5}{13} + 1 - 2 \times 3\frac{5}{2} \div 2\frac{3}{4}] \\ = [(\frac{13}{6}) \text{ of } \frac{18}{13} + 1 - 2 \times \frac{11}{2} \times \frac{4}{11}] \\ = [3 + 1 - 4] = 0$$

$$\text{Sol.45.(b)} \frac{(4x^2 + 9y^2 - 12xy)}{(3y - 2x)}$$

$$\frac{(2x)^2 + (3y)^2 - 2 \times 2x \times 3y}{(3y - 2x)}$$

$$\frac{(3y - 2x)^2}{(3y - 2x)} = 3y - 2x$$

Sol.46.(d)

$$25 \div 5 + 3 \text{ of } 2 - 4 + 6 \text{ of } 3 \div 2 - 8 + 4 \\ = 5 + 6 - 4 + 18 \div 2 - 8 + 4 \\ = 7 + 9 - 8 + 4 = 12$$

Sol.47.(b)

$$= 0.07 \times 0.28 \div 0.04 + 0.64 - 1.64 \div 0.04 \\ = 0.07 \times 7 + 0.64 - 41 \\ = 0.49 - 40.36 = -39.87$$

Sol.48.(c) $11 - \{(3 \div 4) \text{ of } 8 - 4 - 8\}$

$$\Rightarrow 11 - \{6 - (-4)\} \Rightarrow 11 - 10 = 11 - 10 = 1$$

Sol.49.(a) $133 \div 19 \times 3 - 48 \times 2 \div 24 +$

$$841 \div [(45 - 16)]$$

$$= 7 \times 3 - 48 \times \frac{2}{24} + 841 \div 29$$

$$= 21 - 4 + 29 = 46$$

$$\text{Sol.50.(d)} 9 \div [1 + \{4 \times (\frac{5}{6} - \frac{1}{3} + \frac{1}{2})\}]$$

$$9 \div [1 + \{4 \times (\frac{5}{6} - \frac{2+3}{6})\}]$$

$$9 \div [1 + \{4 \times (\frac{5}{6} - \frac{5}{6})\}]$$

$$9 \div [1 + \{4 \times 0\}] \Rightarrow 9 \div 1 = 9$$

Sol.51.(d) $9 \times N \text{ of } 4 \div 4 - 15 \times 4 = 12$

$$9 \times 4N \div 4 - 60 = 12$$

$$9N - 60 = 12 \Rightarrow 9N = 72 \Rightarrow N = 8$$

$$\text{Sol.52.(a)} \frac{3(16^3 - 6^3)}{(16^2 + 6^2 + Q)} = 30$$

$$\frac{3(4096 - 216)}{(256 + 36 + Q)} = 30$$

$$\frac{3(3880)}{(292 + Q)} = 30 \Rightarrow 388 = 292 + Q$$

$$Q = 388 - 292 = 96$$

Sol.53.(b) $(60 + 64 \div 4 \text{ of } 4) \times [1000 \div \{(10 \text{ of } (3 + 2)) \times 2 \text{ of } 5\}] - 28 + (100 \times 3 \text{ of } 3)$

$$= (60 + 64 \div 16) \times [1000 \div \{10 \text{ of } 5 \times 10\}] - 28 + (100 \times 9)$$

$$= (60 + 4) \times [1000 \div 500] - 28 + 900$$

$$= 64 \times 2 - 28 + 900$$

$$= 128 - 28 + 900 = 1000$$

Sol.54.(b) $626 \times 2912 \div 8138$

$$= 626 \times \frac{2912}{8138} = 626 \times \frac{112}{313}$$

$$= 2 \times 112 = 224$$

$$\text{Sol.55.(d)} \frac{4x^2 - 49y^2}{2x + 7y} + 7y - 2x$$

$$= \frac{(2x)^2 - (7y)^2}{2x + 7y} + 7y - 2x$$

$$\frac{(2x + 7y)(2x - 7y)}{2x + 7y} + 7y - 2x$$

$$= 2x - 7y + 7y - 2x = 0$$

$$\text{Sol.56.(b)} \frac{7 + 8 \times 7 \div 8 + 7 \div 7 \div 7}{3 + 3 \times 3 \div 3 + 5 \div 5 \div 5}$$

$$\frac{7 + 7 + 1 \div 7}{3 + 3 \times 1 + 1 \div 5} = \frac{14 + \frac{1}{7}}{6 + \frac{1}{5}} =$$

$$\frac{\frac{99}{7}}{\frac{31}{5}} = \frac{99}{7} \times \frac{5}{31} = \frac{495}{217} = 2\frac{61}{217}$$

$$\text{Sol.57.(c)} \frac{4}{7} \times 4\frac{1}{2} \div 5\frac{1}{3} \text{ of } 2\frac{1}{2} - (7\frac{7}{8}$$

$$\div 5\frac{1}{9} \text{ of } 9\frac{9}{20}) + \frac{1}{2}$$

$$= \frac{4}{7} \times \frac{9}{2} \div \frac{16}{3} \text{ of } \frac{5}{2} - (\frac{63}{8} \div \frac{46}{9} \text{ of } \frac{189}{20}) + \frac{1}{2}$$

$$= \frac{4}{7} \times \frac{9}{2} \div \frac{80}{6} - (\frac{63}{8} \div \frac{483}{10}) + \frac{1}{2}$$

$$= \frac{4}{7} \times \frac{9}{2} \times \frac{6}{80} - (\frac{63}{8} \times \frac{10}{483}) + \frac{1}{2}$$

$$= \frac{27}{140} - (\frac{315}{1932}) + \frac{1}{2} = \frac{27}{140} - (\frac{105}{644}) + \frac{1}{2}$$

$$= \frac{97}{140} - \left(\frac{105}{644}\right) = \frac{97 \times 644 - 105 \times 140}{140 \times 644}$$

$$= \frac{62468 - 14700}{140 \times 644}$$

$$= \frac{47768}{140 \times 644} = \frac{4 \times 2 \times 7 \times 853}{140 \times 644} = \frac{853}{1610}$$

Sol.58.(c) $(3 - 3 \times 3 + 3 \div 3 + 3 \times 5) \times 2$ of $5 + (2 + 2 \div 2 + 2 \times 2 - 2)$

$$= (3 - 9 + 1 + 15) \times 10 + (2 + 1 + 4 - 2)$$

$$= (19 - 9) \times 10 + (7 - 2)$$

$$= (10) \times 10 + (5) = 100 + 5 = 105$$

Sol.59.(a)

$$\{(3^3 + 2^3) \times \frac{22}{7}\} \times \{(6^2 + 3^2 - 4^2) \div (14 \frac{1}{2})\}$$

$$= \{(27 + 8) \times \frac{22}{7}\} \times \{(36 + 9 - 16) \div (\frac{29}{2})\}$$

$$= \{35 \times \frac{22}{7}\} \times \{29 \div (\frac{29}{2})\}$$

$$= \{110\} \times \{29 \times \frac{2}{29}\} = \{110 \times 2\} \Rightarrow 220$$

Sol.60.(d) $[40 - 48 \div 4 \text{ of } 3] \div \{(6 + 4) - 4$
of $4 + 4 + (6 - 4) + (4 \times 4) \div 4\} \times 3$

$$= [40 - 48 \div 12] \div \{10 - 16 + 4 + 2 + 4\} \times 3$$

$$= 36 \div 4 \times 3 = 27$$

Sol.61.(d) $8 - 5 \div 2^3 \times 3 + (7 \div 7 \times 7) \div 7 +$
 $(3 \times 6 \div \frac{1}{3}) \times \frac{1}{3^3}$

$$= 8 - 5 \div 8 \times 3 + (1 \times 7) \div 7 + (3 \times 6 \times 3)$$

$$\times \frac{1}{27}$$

$$= 8 - 5 \times \frac{1}{8} \times 3 + (7) \times \frac{1}{7} + (54) \times \frac{1}{27}$$

$$= 8 - \frac{15}{8} + 1 + 2 = 11 - \frac{15}{8} = \frac{73}{8}$$

Sol.62.(b)

$$10 + 3 \frac{3}{5} \div [2 \frac{2}{3} \div \{7 \frac{1}{6} - (2 \frac{1}{2} + \frac{2}{3})\}]$$

$$= 10 + \frac{18}{5} \div [\frac{8}{3} \div \{\frac{43}{6} - (\frac{5}{2} + \frac{2}{3})\}]$$

$$= 10 + \frac{18}{5} \div [\frac{8}{3} \div \{\frac{43}{6} - (\frac{15+4}{6})\}]$$

$$= 10 + \frac{18}{5} \div [\frac{8}{3} \div \{\frac{43-19}{6}\}]$$

$$= 10 + \frac{18}{5} \div [\frac{8}{3} \times \{\frac{6}{24}\}]$$

$$= 10 + \frac{18}{5} \div [\frac{2}{3}]$$

$$= 10 + \frac{18}{5} \times \frac{3}{2} = 10 + \frac{27}{5} = \frac{77}{5}$$

Sol.63.(b) $3 \frac{4}{7} - [16 \frac{1}{2} \div \{6^2 - (7 - 2 \frac{1}{2})\}]$

$$= \frac{25}{7} - [\frac{33}{2} \div \{36 - (7 - \frac{5}{2})\}]$$

$$= \frac{25}{7} - [\frac{33}{2} \div \{36 - (\frac{9}{2})\}]$$

$$= \frac{25}{7} - [\frac{33}{2} \div \frac{63}{2}]$$

$$= \frac{25}{7} - [\frac{33}{2} \times \frac{2}{63}] = \frac{25}{7} - [\frac{33}{63}]$$

$$= \frac{25 \times 9}{7 \times 9} - \frac{33}{63}$$

$$= \frac{225 - 33}{63} = \frac{192}{63} = \frac{64}{21}$$

Sol.64.(a) $\frac{\frac{1}{2} \div \frac{3}{4} \text{ of } \frac{5}{6}}{\frac{1}{2} \div \frac{3}{4} \times \frac{5}{6}} + \frac{2 + 2 \times 2}{2 \div 2 \times 2}$

$$= \frac{\frac{1}{2} \div \frac{15}{24}}{\frac{2}{3} \times \frac{5}{6}} + \frac{2 + 4}{2} = \frac{\frac{12}{15}}{\frac{5}{9}} + 3$$

$$= \frac{12 \times 3}{5 \times 5} + 3 = \frac{36}{25} + 3$$

$$= \frac{36 + 75}{25} = \frac{111}{25}$$

Sol.65.(c) $4 \times 7 - 3\{7 \times 4 \div (6 \times 2)\}$

$$= 28 - 3\{7 \times 4 \div 12\}$$

$$= 28 - 3\{\frac{7}{3}\} = 28 - 7 = 21$$

Sol.66.(c) $420 \div [\frac{4}{9} \times (1 + \frac{4}{5}) \times 5] +$
 $[\frac{5}{7} \text{ of } \frac{21}{25} \div \frac{3}{4} \text{ of } \frac{2}{5}]$

$$= 420 \div [\frac{4}{9} \times (\frac{9}{5}) \times 5] + [\frac{3}{5} \div \frac{3}{10}]$$

$$= 420 \div 4 + [\frac{3}{5} \times \frac{10}{3}] = 105 + 2 = 107$$

Sol.67.(c) $\frac{2 \times 2 + 2}{2 \div 2 \times 2} \div \frac{\frac{1}{2} \div \frac{1}{2} \text{ of } \frac{1}{2}}{\frac{1}{2} + \frac{1}{2} \text{ of } \frac{1}{2}}$

$$\frac{4 + 2}{1 \times 2} \div \frac{\frac{1}{2} \div \frac{1}{4}}{\frac{1}{2} + \frac{1}{4}} = 3 \div \frac{\frac{1}{2} \times 4}{\frac{6}{8}}$$

$$3 \div \frac{2 \times 8}{6} = 3 \times \frac{6}{16} = \frac{9}{8} = 1 \frac{1}{8}$$

Sol.68.(d) $(24 \div 3 \text{ of } \frac{1}{6}) \times (\frac{4}{5} \div \frac{9}{8} \times$
 $\frac{3}{5}) \div (\frac{4}{7} \div \frac{3}{14} \text{ of } \frac{3}{14})$

$$= (24 \div \frac{1}{2}) \times (\frac{4}{5} \times \frac{8}{9} \times \frac{3}{5}) \div (\frac{4}{7} \div \frac{9}{196})$$

$$= (24 \times 2) \times (\frac{32}{75}) \div (\frac{4}{7} \times \frac{196}{9})$$

$$= 48 \times (\frac{32}{75}) \div (\frac{112}{9})$$

$$= 48 \times (\frac{32}{75}) \times (\frac{9}{112})$$

$$= 48 \times (\frac{2}{25}) \times (\frac{3}{7}) = \frac{288}{175}$$

Sol.69.(b)

$$= \frac{5 + 7 \times 5 \div 8 \text{ of } 5 + 3 \div 3 \times 5}{6 \div 6 \text{ of } 4 + 9 \times 9 \div 3 \times 2 \times 2}$$

$$= \frac{5 + 7 \times 5 \div 40 + 1 \times 5}{6 \div 24 + 27 \times 2 \times 2}$$

$$= \frac{\frac{7}{8} + 10}{\frac{1}{4} + 108} = \frac{\frac{7+80}{8}}{\frac{1+432}{4}}$$

$$= \frac{\frac{87}{8}}{\frac{433}{4}} = \frac{87 \times 4}{8 \times 433} = \frac{87}{866}$$

Sol.70.(b) $11 \frac{1}{4} \div 3 \frac{3}{5} \times 6 \frac{2}{3} + \frac{1}{6}$

$$= \frac{45}{4} \div \frac{18}{5} \times \frac{20}{3} + \frac{1}{6}$$

$$= \frac{45}{4} \times \frac{5}{18} \times \frac{20}{3} + \frac{1}{6}$$

$$= \frac{125}{6} + \frac{1}{6} = \frac{126}{6} = 21$$

Sol.71.(c)

$$[3^3 \{(40 \times 5) + (7 \times 6)\} \div 11] + (2^3 \times 15 \div 24)$$

$$= [27 \{(200 + 42) \div 11\} + 5]$$

$$= [27 \{242\} \div 11] + 5$$

$$= [27 \times 22] + 5 = 594 + 5 = 599$$

Sol.72.(d) Given expression :-

$$47 \div (10 \times 5) + (4 - 5) - 6$$

Replacing the signs as per the directions given, we get ;

$$47 \times (10 \div 5) - (4 + 5) + 6$$

$$= 47 \times 2 - 9 + 6 = 94 - 3 = 91$$

Sol.73.(a) $\frac{[\frac{1}{4} \div \frac{1}{4} \times \frac{1}{4}] \times \frac{1}{4}}{[\frac{1}{8} + \frac{1}{8} \times \frac{1}{8}]} + (1 - \frac{4}{9})$

$$= \frac{\frac{1}{4} \times \frac{1}{4}}{\frac{1}{8} + \frac{1}{64}} + \frac{5}{9} = \frac{\frac{1}{16}}{\frac{9}{64}} + \frac{5}{9}$$

$$= \frac{1}{16} \times \frac{64}{9} + \frac{5}{9} = \frac{4}{9} + \frac{5}{9} = \frac{9}{9} = 1$$

Sol.74.(d) $3 \frac{1}{5} \text{ of } 2 \frac{3}{4} \div \frac{11}{3} + (2 \frac{1}{5} \div 1$
 $\frac{2}{5} \text{ of } 1 \frac{3}{7} - 2 \frac{1}{3})$

$$= \frac{16}{5} \text{ of } \frac{11}{4} \div \frac{11}{3} + (\frac{11}{5} \div \frac{7}{5} \text{ of } \frac{10}{7} -$$

$$\frac{7}{3}) = \frac{12}{5} + (\frac{11}{10} - \frac{7}{3})$$

$$= \frac{12}{5} - \frac{37}{30} = \frac{35}{30} = \frac{7}{6} = 1 \frac{1}{6}$$

Sol.75.(b) $[18 - 3 \times \{3 - 3 \times (3 \div \frac{1}{3} - 3) + 3$
 $\div \frac{1}{3} + 3\} - 4 \times 2]$

$$= [18 - 3 \times \{3 - 3 \times 6 + 12\} - 8]$$

$$= [18 - 3 \times (-3) - 8] = 18 + 9 - 8 = 19$$

Sol.76.(c) $12 + 24 \div 2 - 7 \times 30 \div 6 + (5 +$
 $4) \times 8 + 8 \div 8 - 8$

$$= 12 + 12 - 7 \times 5 + (9) \times 8 + 1 - 8$$

$$= 12 + 12 - 35 + 72 + 1 - 8 = 97 - 43 = 54$$

Sol.77.(a) $\frac{(143 \div 13 \times x - 3 \times 3)}{(6^2 - 7 \times 5 + x^2)} = 1$

$$\Rightarrow \frac{11x - 9}{36 - 35 + x^2} = 1 \Rightarrow 11x - 9 = 1 + x^2$$

$$\Rightarrow x^2 - 11x + 10 = 0 \Rightarrow x^2 - 10x - x + 10 = 0$$

$$\Rightarrow x(x - 10) - 1(x - 10) = 0$$

$$\Rightarrow (x - 10)(x - 1) = 0 \Rightarrow x = 10 \text{ or } 1$$

$$\Rightarrow x = 1 \text{ (from option)}$$

Sol.78.(b)

$$\frac{12 + 3 \frac{2}{3} \div [1 \frac{3}{4} \div \{5 \frac{1}{4} \times (3 \frac{3}{5} \div 4 \frac{1}{2})\}]}{1 \frac{1}{3} \times 5 \frac{1}{5}}$$

$$\begin{aligned}
 &= \frac{12 + \frac{11}{3} \div \left[\frac{7}{4} \div \left\{ \frac{21}{4} \times \left(\frac{18}{5} \div \frac{9}{2} \right) \right\} \right]}{\frac{4}{3} \times \frac{26}{5}} \\
 &= \frac{12 + \frac{11}{3} \div \left[\frac{7}{4} \div \frac{21}{5} \right]}{\frac{104}{15}} \\
 &= \frac{12 + \frac{11}{3} \div \frac{5}{12}}{\frac{104}{15}} = \frac{12 + \frac{44}{5}}{\frac{104}{15}} \\
 &= \frac{\frac{104}{5}}{\frac{104}{15}} = \frac{104}{5} \times \frac{15}{104} = 3
 \end{aligned}$$

Sol.79.(c) Formula used :

$$\begin{aligned}(a + b + c)^2 &= a^2 + b^2 + c^2 + 2(ab + bc + ca) \\ 25 \times 25 + 9 \times 9 + 6 \times 6 + 2 \times 25 \times 9 + 2 \times 9 \\ &\quad \times 6 + 2 \times 25 \times 6 \\ &= 25^2 + 9^2 + 6^2 + 2 \times 25 \times 9 + 2 \times 9 \times 6 + 2 \\ &\quad \times 25 \times 6 = (25 + 9 + 6)^2 = 1600\end{aligned}$$

Sol.80.(b)

$$\begin{aligned} 1 - \frac{7}{1 - \frac{1}{1 + \frac{7}{1-7}}} &= 1 - \frac{7}{1 - \frac{1}{1 - \frac{7}{6}}} \\ &= 1 - \frac{7}{1 - \frac{1}{-\frac{1}{6}}} = 1 - \frac{7}{1+6} = 1 - 1 = 0 \end{aligned}$$

Sol.81.(a) $9^{18} \div 3^{14}$ of $27^3 \times \sqrt{6561}$
 $= (3^2)^{18} \div 3^{14}$ of $(3^3)^3 \times 3^4$
 $= 3^{36} \div 3^{23} \times 3^4 \Rightarrow 3^{13} \times 3^4 = 3^{17}$

Sol.82.(c) $3 + 2 \times [7 + (5 + 7) \div 2]$
 $= 3 + 2 \times [7 + 12 \div 2]$
 $= 3 + 2 \times 13 = 3 + 26 = 29$

Sol.83.(a) $\frac{92736}{(274 + 830) \text{ of } \frac{7}{4}} + 513$

$= \frac{92736}{1104 \text{ of } \frac{7}{4}} + 513 = \frac{92736}{276 \times 7} + 513$

$\Rightarrow 48 + 513 = 561$

$$\begin{aligned}\text{Sol.84.(d)} \quad & \frac{(525 + 175)^2 - (525 - 175)^2}{(525 \times 175)} \\ &= \frac{(525 + 175 + 525 - 175)(525 + 175 - 525 + 175)}{(525 \times 175)} \\ &= \frac{(1050) \times (350)}{(525 \times 175)} = 4\end{aligned}$$

Short Trick : Using the formula,

$$\frac{(a+b)^2 - (a-b)^2}{(525+175)^2 - (525-175)^2} = \frac{4ab}{(525 \times 175)}$$
$$= \frac{4 \times 525 \times 175}{(525 \times 175)} = 4$$

Sol.85.(b) $9 \times 9 + 9 \div 9 - 9 \times 9 + 9 + 9 \times 9$
 $- 9 - 9 \times 9$
 $= 81 + 1 - 81 + 9 + 81 - 9 - 81$
 $= 1 + 9 - 9 = 1$

Sol.86.(a) $(4^2)^2 + 4 \times 4 \div 4 - 4$
 $= 256 + 4 \times 1 - 4 = 256 + 4 - 4 = 256$ or 4^4

Sol.87.(d) $1 + \frac{1}{1 + \frac{1}{5 + \frac{1}{2 + \frac{1}{3}}}}$

$$= 1 + \frac{1}{1 + \frac{1}{5 + \frac{3}{7}}} = 1 + \frac{1}{1 + \frac{7}{38}}$$
$$= 1 + \frac{38}{45} = 1 + \frac{38}{45} = 1\frac{38}{45}$$

Sol.88.(c)

$$\begin{aligned}x[-5\{-4(-a)\}] + 6[-3\{-3(-a)\}] &= 6a \\ \Rightarrow x[-5\{4a\}] + 6[-3\{3a\}] &= 6a \\ \Rightarrow x[-20a] + 6[-9a] &= 6a \\ \Rightarrow x[-20a] &= 6a + 54a \\ \Rightarrow x[-20a] &= 60a \Rightarrow x = \frac{60a}{-20a} = -3\end{aligned}$$

Sol.89.(a) $20 \div 5 \text{ of } 4 \times 8 + 6 \div 24 \text{ of } 3 \times$
 $48 - 6 \div 18 \times 9$
 $= 20 \div 20 \times 8 + 6 \div 72 \times 48 - \frac{1}{3} \times 9$

$$= 1 \times 8 + \frac{1}{12} \times 48 - \frac{1}{3} \times 9 = 8 + 4 - 3 = 9$$

Sol.90.(c) $64 \div [44 - (8 \text{ of } 3 - 16) \div 4 \times 20]$
 $= 64 \div [44 - (24 - 16) \div 4 \times 20]$
 $= 64 \div [44 - 8 \div 4 \times 20]$
 $= 64 \div [44 - 2 \times 20] = 64 \div [44 - 40] = 16$

Sol.91.(b) $855 \div 15 - k + 32 \times 5 = 1152 \div 16 \times 111 \div 37$
 $\Rightarrow 57 - k + 160 = 72 \times 3$
 $\Rightarrow 57 - k + 160 = 216$
 $\Rightarrow -k = 216 - 217 \Rightarrow k = 1$

Sol.92.(c)
 $1000 \times 100 - 1000 + 4000 \div 200$
 $= 100000 - 1000 + 20$
 $= 99000 + 20 = 99020$

Sol.93.(b) $15\% \text{ of } 1.5 + 0.15 \div 0.1 + 0.01$
of $1.1 \times 0.5 + 2.5 \times 25$.
 $= 0.225 + 1.5 + 0.0055 + 62.5 = 64.2305$

Sol.94.(c) $(6 + 6 \text{ of } 5 \div 5) \div 6 \text{ of } 5 + (24 \div 2 \text{ of } 3 - 6 \text{ of } 5 + 4)$
 $= (6 + 30 \div 5) \div 30 + (24 \div 6 - 30 + 4)$
 $= (6 + 6) \div 30 + (4 - 30 + 4)$
 $= 12 \div 30 - 22$
 $= \frac{2}{5} - 22 = \frac{-108}{5} = -21\frac{3}{5}$

Sol.95.(b)
 $5055 - (1002 \div 20.04) = 5055 - 50 = 5005$

Sol.96.(d)
 $56 \div 4 + p \times 3 - 24 \div 12 + 4 = 21 \div 3 + 3$
 $= 14 + p \times 3 - 2 + 4 = 7 + 3$
 $= 16 + p \times 3 = 10 = p \times 3 = -6 \Rightarrow p = -2$

Sol.97.(b)
 $22.0625 - [15 + 0.25 \times (4.78 - 1.5 \times 3.02)]$
 $= 22.0625 - [15 + 0.25 \times (4.78 - 4.53)]$
 $= 22.0625 - [15 + 0.25 \times 0.25]$
 $= 22.0625 - [15 + 0.0625]$
 $= 22.0625 - 15.0625 = 7.0$

Sol.98.(a) $(4.2 \text{ of } 2.5 \div 3.5) - (3.6 \div 2.4 + 2.5 \text{ of } 3 \div 7.5) + (6 - 3 \text{ of } 2 + 4 \text{ of } 3 \div 6)$
 $= (10.5 \div 3.5) - (1.5 + 7.5 \div 7.5) + (6 - 6 + 12 \div 6)$
 $= (3) - (1.5 + 1) + (2) \Rightarrow 3 - 2.5 + 2 = 2.5$

Sol.99.(c) $\frac{13 \div \{6 - 6 \div (7 - 9) \times 8 + 9\}}{5 + 5 \times 5 \div 5 \text{ of } 5}$
 $= \frac{13 \div \{6 - 6 \div (-2) \times 8 + 9\}}{5 + 5 \times 5 \div 25}$
 $= \frac{13 \div \{6 + 3 \times 8 + 9\}}{5 + 1}$
 $= \frac{13 \div \{6 + 24 + 9\}}{6} = \frac{13 \div 39}{6} = \frac{1}{18}$

$$\begin{aligned} \text{Sol.100.(c)} \quad & \left(2\frac{1}{7} + 6\frac{1}{7} \right) \div \frac{9}{7} \\ & \frac{1}{\frac{1}{9}} \div \left\{ \frac{7}{9} + \left(\frac{4}{9} \div \frac{4}{3} \right) \right\} \\ & \left(\frac{15}{7} + \frac{43}{7} \right) \div \frac{9}{7} \\ & = \frac{10}{9} \div \left\{ \frac{7}{9} + \left(\frac{4}{9} \times \frac{3}{4} \right) \right\} \\ & \left(\frac{58}{7} \right) \times \frac{7}{9} = \frac{58}{9} \\ & = \frac{10}{9} \div \left\{ \frac{7}{9} + \frac{1}{3} \right\} = \frac{10}{9} \div \frac{10}{9} \\ & \frac{58}{9} \\ & = \frac{10}{9} \times \frac{9}{10} = \frac{58}{9} \end{aligned}$$

$$\begin{aligned} \text{Sol. 101. (a)} \\ & \frac{(275 \times 275 \times 275 + 725 \times 725 \times 725)}{(275 \times 275 + 725 \times 725 - 725 \times 275)} \\ & \times \frac{(275 \times 275 \times 275 - 725 \times 725 \times 725)}{(275 \times 275 + 725 \times 725 + 725 \times 275)} \\ & = \frac{(275)^3 + (725)^3}{(275)^2 + (725)^2 - 725 \times 275} \\ & \times \frac{(275)^3 - (725)^3}{(275)^2 + (725)^2 + 725 \times 275} \\ & = \frac{(275 + 725) [(275)^2 + (725)^2 - 275 \times 725]}{(275)^2 + (725)^2 - 725 \times 275} \\ & \times \frac{(275 - 725) [(275)^2 + (725)^2 + 275 \times 725]}{(275)^2 + (725)^2 + 725 \times 275} \\ & = (275 + 725) \times (275 - 725) \\ & = 1000 \times (-450) = -4,50,000 \end{aligned}$$

Sol.102.(b) $\frac{5x+3y}{5x-3y} = \frac{7}{3}$
 $15x+9y=35x-21y$
 $30y=20x \Rightarrow \frac{x}{y} = \frac{30}{20} = \frac{3}{2}$ or $3:2$

Sol.103.(a) $\frac{1}{2^2} + \frac{1}{2^3} \div \left(\frac{1}{2} + \frac{1}{2} \div 1 \right) + 3$

$$\begin{aligned} & \frac{1}{3} \div \frac{5}{2} \times \frac{3}{4} \div 6 \frac{2}{3} \times \frac{7}{6} \\ &= \frac{1}{4} + \frac{1}{8} \div \left(\frac{1}{2} + \frac{1}{2} \right) + \frac{10}{3} \times \frac{2}{5} \times \frac{3}{4} \times \\ & \frac{3}{20} \times \frac{7}{6} = \frac{1}{4} + \frac{1}{8} + \frac{7}{40} \\ &= \frac{10+5+7}{40} = \frac{22}{40} = \frac{11}{20} \end{aligned}$$

Sol.104.(d)
 $12.3 - [7 + 0.8 \text{ of } (5.6 - 3.8 \times 1.03)]$
 $= 12.3 - [7 + 0.8 \text{ of } (5.6 - 3.914)]$
 $= 12.3 - [7 + 1.3488]$

$$= 12.3 - 8.3488 = 3.9512$$

$$\text{Sol.105.(d)} \quad x + \frac{1}{1 + \frac{1}{3 + \frac{1}{2}}} = 1$$

$$\Rightarrow x + \frac{1}{1 + \frac{2}{7}} = 1 \Rightarrow x + \frac{7}{9} = 1$$

$$\Rightarrow x = 1 - \frac{7}{9} = \frac{2}{9}$$

Sol.106.(c)

$$55\frac{1}{3} \div [3\frac{1}{5} + \{2\frac{1}{2} - (2\frac{1}{3} + \frac{3}{5})\}]$$

$$= \frac{166}{3} \div [\frac{16}{5} + \{2 - (\frac{7}{3} + \frac{3}{5})\}]$$

$$= \frac{166}{3} \div [\frac{16}{5} + \{2 - \frac{44}{15}\}]$$

$$= \frac{166}{3} \div \frac{83}{30} = \frac{166}{3} \times \frac{30}{83} = 20$$

Sol.107.(b)

$$8.16 \times 5.35 + 17.9 - 19.5 \text{ of } \frac{1}{15} + \frac{1}{5}$$

$$= 43.656 + 17.9 - 19.5 \times \frac{1}{15} + \frac{1}{5}$$

$$= 61.556 - 1.3 + 0.2 = 60.456$$

$$\text{Sol.108.(c)} \quad \frac{53 - \frac{2}{7} \text{ of } 63 - 9}{72 - \frac{2}{5} \text{ of } (96 - 31)}$$

$$= \frac{53 - 2 \times 9 - 9}{72 - \frac{2}{5} \text{ of } 65} = \frac{53 - 18 - 9}{72 - 2 \times 13}$$

$$= \frac{26}{46} = \frac{13}{23}$$

Sol.109.(d) $25^3 - 75^3 + 50^3$

$$= (25 - 75)(25^2 + 75^2 + 25 \times 75) + 50^3$$

$$= (-50)(625 + 5625 + 1875) + 125000$$

$$= (-50) \times 8125 + 125000$$

$$= -406250 + 125000 = -281250$$

Short trick:-

$$a = 25, b = -75, c = 50$$

$$\text{If } a + b + c = 0, \text{ then } a^3 + b^3 + c^3 = 3abc$$

$$\text{So, } 25^3 - 75^3 + 50^3$$

$$= -3 \times 25 \times 75 \times 50 = -281250$$

Sol.110.(c) We know

$$(a + b + c)^3$$

$$= a^3 + b^3 + c^3 + 3(a+b)(b+c)(c+a)$$

$$\text{Here, } a = 0.2$$

$$b = 0.3 \Rightarrow c = 0.5$$

Hence,

$$(0.2)^3 + (0.3)^3 + (0.5)^3 + 3 \times 0.5 \times 0.7 \times 0.8$$

$$= (0.2 + 0.3 + 0.5)^3 \Rightarrow (1.0)^3 = 1.000$$

Sol.111.(a)

$$\frac{0.7^2 + (0.4 + 0.2)(0.7) + (0.4 \times 0.2)}{(0.7 + 0.4)}$$

$$= \frac{0.49 + 0.42 + 0.08}{1.1} = \frac{0.99}{1.1} = 0.9$$

Sol.112.(d) Formula used :

$$(a - b)^3 = a^3 - b^3 - 3a^2b + 3ab^2$$

$$\frac{(8.25)^3 - 3 \times (8.25)^2 \times 0.25 + 3 \times 8.25 \times (0.25)^2 - (0.25)^3}{4 \times 4 \times 4}$$

$$= \frac{(8.25 - 0.25)^3}{64} = \frac{512}{64} = 8$$

Sol.113.(d) Formula used :

$$a^3 + b^3 + c^3 - 3abc$$

$$= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\frac{(2.3)^3 + (15.9)^3 + (3.7)^3 - 3 \times 2.3 \times 15.9 \times 3.7}{(2.3)^2 + (15.9)^2 + (3.7)^2 - 2.3 \times 15.9 - 15.9 \times 3.7 - 3.7 \times 2.3}$$

$$\frac{(2.3+15.9+3.7)(2.3^2+15.9^2+3.7^2-2.3 \times 15.9-15.9 \times 3.7-3.7 \times 2.3)}{(2.3)^2+(15.9)^2+(3.7)^2-2.3 \times 15.9-15.9 \times 3.7-3.7 \times 2.3}$$

$$= (2.3 + 15.9 + 3.7) = 21.9$$

Sol.114.(c) Identity used:

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$\frac{12.43 \times 12.43 \times 12.43 + 19.57 \times 19.57 \times 19.57}{12.43 \times 12.43 - 12.43 \times 19.57 + 19.57 \times 19.57}$$

$$= \frac{12.43^3 + 19.57^3}{12.43^2 - 12.43 \times 19.57 + 19.57^2}$$

$$=$$

$$\frac{(12.43 + 19.57)(12.43^2 - 12.43 \times 19.57 + 19.57^2)}{(12.43^2 - 12.43 \times 19.57 + 19.57^2)}$$

$$= 32$$

Sol.115.(d) Concept used :

$$(a - b)^3 = a^3 - b^3 - 3a^2b + 3ab^2$$

$$= 15 \times 15 \times 15 + 3 \times 15 \times 12 \times 12 - 3 \times 15 \times 15 \times 12 - 12 \times 12 \times 12$$

$$= 15^3 + 3 \times 15 \times 12^2 - 3 \times 15^2 \times 12 - 12^3$$

$$= (15 - 12)^3 = 3^3 = 27$$

Sol.116.(b) Formula used :

$$(a - b)(a + b) = a^2 - b^2$$

$$\frac{((32)^4 - 1)}{33 \times 31 \times (2^{10} + 1)}$$

$$= \frac{((2^5)^4 - 1)}{(2^5 + 1)(2^5 - 1)(2^{10} + 1)}$$

$$= \frac{2^{20} - 1}{(2^{10} - 1)(2^{10} + 1)} = \frac{2^{20} - 1}{2^{20} - 1} = 1$$

Sol.117.(c)

$$\sqrt[3]{0.08 \times 0.8 \times p} = 0.008 \times 0.8 \times \sqrt[3]{q}$$

$$8 \times 8 \times 10^{-3} \times p = (8 \times 8 \times 10^{-4})^3 \times q$$

$$8^2 \times 10^{-3} \times p = 8^6 \times 10^{-12} \times q$$

$$\frac{p}{q} = 8^4 \times 10^{-9}$$

$$\text{Sol.118.(b)} \quad (\sqrt[3]{343})^3 \times (\sqrt[3]{216})^3$$

$$= \sqrt[3]{343} \times \sqrt[3]{216} = 7 \times 6 = 42$$

Sol.119.(a)

$$\frac{\sqrt{1.24} \times \sqrt{2.79}}{\sqrt{2.64} \times \sqrt{5.94}} = \sqrt{\frac{1.24 \times 2.79}{2.64 \times 5.94}}$$

$$= \sqrt{\frac{31 \times 31}{66 \times 66}} = \frac{31}{66}$$

Sol.120.(c)

$$= \frac{0.04}{0.05} \text{ of } \frac{(3\frac{1}{3} - 2\frac{1}{2}) \div \frac{1}{2} \text{ of } 1\frac{1}{4}}{\frac{1}{3} + \frac{1}{5} \text{ of } \frac{1}{9}}$$

$$= \frac{4}{5} \text{ of } \frac{(\frac{10}{3} - \frac{5}{2}) \div \frac{1}{2} \text{ of } \frac{5}{4}}{\frac{1}{3} + \frac{1}{5} \text{ of } \frac{1}{9}}$$

$$= \frac{4}{5} \text{ of } \frac{(\frac{5}{6}) \div \frac{5}{8}}{\frac{1}{3} + \frac{1}{45}}$$

$$= \frac{4}{5} \text{ of } \frac{(\frac{4}{3})}{\frac{16}{45}} = \frac{4}{5} \times \frac{15}{4} = 3$$

Sol.121.(d) $(48 \div 72 \times 3) - [15 \div 8 \times (40 - 32) - 10] + 2P = 6 \div 2$

$$\Rightarrow (48 \div 72 \times 3) - [15 \div 8 \times 8 - 10] + 2P = 6 \div 2$$

$$\Rightarrow (48 \div 72 \times 3) - [\frac{15}{8} \times 8 - 10] + 2P = 6 \div 2$$

$$\Rightarrow (\frac{2}{3} \times 3) - [15 - 10] + 2P = 6 \div 2$$

$$\Rightarrow 2 - 5 + 2P = 3 \Rightarrow P = 3$$

Sol.122.(c)

$$1\frac{1}{2} \div 3\frac{1}{4} + \frac{1}{2} \div \frac{13}{14} + \frac{1}{5}$$

$$\frac{1}{5} \times 3\frac{1}{2} - \frac{1}{3} \div 1\frac{3}{4} \times 3\frac{1}{2}$$

$$= \frac{3}{2} \div \frac{13}{4} + \frac{1}{2} \div \frac{13}{14} + \frac{1}{5}$$

$$= \frac{1}{5} \times \frac{7}{2} - \frac{1}{3} \div \frac{7}{4} \times \frac{7}{2}$$

$$= \frac{3}{2} \times \frac{4}{13} + \frac{1}{2} \times \frac{14}{13} + \frac{1}{5}$$

$$= \frac{1}{5} \times \frac{7}{2} - \frac{1}{3} \times \frac{4}{7} \times \frac{7}{2}$$

$$= \frac{\frac{6}{13} + \frac{7}{13} + \frac{1}{5}}{\frac{7}{10} - \frac{2}{3}} = \frac{\frac{6}{5}}{\frac{1}{30}} = 36$$

Sol.123.(b)

$$b + 2a - [(3b + a) - (2a + b) + 2a] - a$$

$$= b + 2a - [3b + a - 2a - b + 2a] - a$$

$$= b + 2a - [2b + a] - a$$

$$= b + 2a - 2b - a - a = -b$$

Sol.124.(d) $71 - (-3)\{-2 - (8 - 3)\} + 3\{5 + (-2)(-1)\}$

$$= 71 - (-3)\{-2 - 5\} + 3\{5 + 2\}$$

$$= 71 - (-3)\{-7\} + 3\{7\}$$

$$= 71 - 21 + 21 = 71$$

$$\text{Sol.125.(c)} \quad 3\frac{1}{2} - [2\frac{1}{4} \div \{1\frac{1}{4} - \frac{3}{4}(1\frac{1}{2} - \frac{1}{2} - 2\frac{1}{2})\}]$$

$$= \frac{7}{2} - [\frac{9}{4} \div \{\frac{5}{4} - \frac{3}{4}(\frac{3}{2} - \frac{1}{2} - \frac{5}{2})\}]$$

$$= \frac{7}{2} - [\frac{9}{4} \div \{\frac{5}{4} - \frac{3}{4}(-\frac{3}{2})\}]$$

$$= \frac{7}{2} - [\frac{9}{4} \div \{\frac{5}{4} + \frac{9}{8}\}]$$

$$= \frac{7}{2} - \frac{18}{19} = \frac{97}{38} = 2\frac{21}{38}$$

Sol.126.(d)

$$\begin{aligned} & \left[\frac{1}{5} \div \frac{1}{5} \times \frac{1}{5} \right] \times \frac{1}{5} + \left(1 - \frac{6}{25} \right) \\ & \left[\frac{1}{9} + \frac{1}{9} \times \frac{1}{9} \right] \times \frac{1}{10} \\ & = \left[\frac{1 \times \frac{1}{5}}{\frac{1}{9} + \frac{1}{81}} \right] \times \frac{1}{10} + \left(\frac{19}{25} \right) \\ & = \frac{\frac{1}{25}}{\left[\frac{10}{81} \right] \times \frac{1}{10}} + \left(\frac{19}{25} \right) \\ & = \frac{1}{25} \times \frac{81}{1} + \left(\frac{19}{25} \right) = \frac{81+19}{25} = \frac{100}{25} = 4 \end{aligned}$$

Sol.127.(b) $87 \times 256 + 16 - 77 \div 1359$

After changing sign ,
 $87 + 256 \div 16 \times 77 - 1359$
 $= 87 + 16 \times 77 - 1359$
 $= 87 + 1232 - 1359 = -40$

Sol.128.(d) $\sqrt{a} = 3b \Rightarrow b^2 = \frac{a}{9}$,

Now, $\frac{a}{b^2} = \frac{9}{1} = 9$

Sol.129.(c)

$$\begin{aligned} & \frac{(0.25)^4 + 2 \times (0.25)^2 + 1 - (0.25)^2}{(0.25)^2 + 0.25 + 1} \\ & = \frac{\{(0.25)^2 + 1\}^2 - (0.25)^2}{(0.25)^2 + 0.25 + 1} \\ & = \frac{\{(0.25)^2 + 1 + 0.25\} \{(0.25)^2 + 1 - 0.25\}}{(0.25)^2 + 0.25 + 1} \\ & = \{(0.25)^2 + 1 - 0.25\} \\ & = \{0.0625 + 1 - 0.25\} = 0.8125 \end{aligned}$$

Sol.130.(b)

$$\begin{aligned} & \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}} = \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}} \times \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} + \sqrt{2}} \\ & = \frac{5 + 2\sqrt{6}}{3 - 2} = 5 + 2 \times 2.45 = 9.9 \end{aligned}$$

Sol.131.(a) Formula used :-

$$\begin{aligned} & a^3 - b^3 = (a - b)(a^2 + b^2 + ab) \\ & \frac{(4.2)^3 - 0.008}{(4.2)^2 + 0.84 + 0.04} \\ & = \frac{(4.2)^3 - (0.2)^3}{(4.2)^2 + 0.84 + 0.04} \\ & = \frac{(4.2 - 0.2)[(4.2)^2 + 4.2 \times 0.2 + (0.2)^2]}{(4.2)^2 + 0.84 + 0.04} \\ & = \frac{4[(4.2)^2 + 0.84 + 0.04]}{(4.2)^2 + 0.84 + 0.04} = 4 \end{aligned}$$

Sol.132.(a) $\frac{10x^2 + 5x + 2xy + y}{5x + y}$

$$\frac{2x(5x + y) + (5x + y)}{5x + y} = \frac{(5x + y)(2x + 1)}{5x + y}$$

$2x + 1 = 2 \times (-1) + 1 = -1$

Sol.133.(d)

$$\begin{aligned} & y + 2x - [y - (y - x + y) - (x + y) + y] - 2y \\ & = y + 2x - [(x - y) - x] - 2y \\ & = y + 2x - [-y] - 2y \\ & = y + 2x + y - 2y = 2x \end{aligned}$$

Sol.134.(c) $\frac{3a+b}{2} - \frac{a-3b}{3} + 2b$

$$\begin{aligned} & = \frac{3(3a+b) - 2(a-3b) + 12b}{6} \\ & = \frac{9a+3b-2a+6b+12b}{6} \\ & = \frac{7a+21b}{6} = \frac{7(a+3b)}{6} \end{aligned}$$

Sol.135.(c) $(a-2b)(b-3a) + (a+b)(a-3b) - (b-3a)(4a-5b)$

$$\begin{aligned} & = -3a^2 - 2b^2 + 7ab + a^2 - 3b^2 - 2ab - (-12a^2 - 5b^2 + 19ab) \\ & = (-3a^2 + a^2 + 12a^2) + (-2b^2 - 3b^2 + 5b^2) + (7ab - 2ab - 19ab) \\ & = 10a^2 - 14ab \end{aligned}$$

Sol.136.(b) $(4x+1)^2 - (4x+3)(4x-1)$

$$\begin{aligned} & = 16x^2 + 1 + 8x - (16x^2 - 4x + 12x - 3) \\ & = 16x^2 + 1 + 8x - 16x^2 + 4x - 12x + 3 \\ & = 4 \end{aligned}$$

Short Tricks :- Put $x = 0$,

$$\begin{aligned} & (4x+1)^2 - (4x+3)(4x-1) \\ & = (1)^2 - (3)(-1) = 4 \end{aligned}$$

Sol.137.(c) $\frac{3y}{1 + \frac{1}{1 + \frac{y}{1-y}}} = 2$

$$\begin{aligned} & \frac{3y}{1 + \frac{1}{\frac{1-y+y}{1-y}}} = \frac{3y}{1 + \frac{1-y}{1}} = \frac{3y}{2-y} = 2 \\ & \Rightarrow 4 - 2y = 3y \Rightarrow y = \frac{4}{5} \end{aligned}$$

Sol.138.(d) $\frac{\sqrt{x}-\sqrt{y}}{\sqrt{x}+\sqrt{y}} = \frac{\sqrt{2}-\sqrt{3}}{\sqrt{2}+\sqrt{3}}$

$$\begin{aligned} & = \frac{\sqrt{2}-\sqrt{3}}{\sqrt{2}+\sqrt{3}} \times \frac{\sqrt{2}-\sqrt{3}}{\sqrt{2}-\sqrt{3}} \\ & = \frac{5-2\sqrt{6}}{2-3} = 2\sqrt{6} - 5 \end{aligned}$$

Sol.139.(b) $52 \times 52 - 48 \times 48$

$$= (52 + 48)(52 - 48) = 100 \times 4 = 400$$

Sol.140.(b) $\frac{17}{30}, \frac{7}{15}, \frac{27}{50}, \frac{11}{20}$

Multiply by 300 in each fraction, we get
 170, 140, 162, 165

Descending order :-

$$\frac{17}{30} > \frac{11}{20} > \frac{27}{50} > \frac{7}{15}$$

Sol.141.(b) $\left(\frac{1}{5}\right) + \left[1\frac{1}{4} - \frac{1}{2}\right] \times \left(\frac{2}{5}\right)$

$$\begin{aligned} & = \frac{1}{5} + \left[\frac{3}{4}\right] \times \left(\frac{2}{5}\right) = \frac{1}{5} + \frac{3}{10} = \frac{5}{10} = 0.5 \\ & 0.5 \text{ lies between } 0.2 \text{ and } 0.6 \end{aligned}$$

Sol.142.(d) $0.5 + 0.55 + 0.555 + 0.5555$

$$\begin{aligned} & = 5(0.1 + 0.11 + 0.111 + 0.1111) \\ & = 5(0.4321) = 2.1605 \end{aligned}$$

Sol.143.(b)

$$\begin{aligned} & \sqrt{[(7^2 + 6^3 + 9^2) \div 2 + 52]} \\ & = \sqrt{[(49 + 216 + 81) \div 2 + 52]} \end{aligned}$$

$$= \sqrt{[346 \div 2 + 52]} = \sqrt{[225]} = 15$$

Sol.144.(c)

$$\begin{aligned} & \sqrt{0.52 \times 0.52 + 0.31 \times 0.31 + 2 \times 0.52 \times 0.31} \\ & = \sqrt{(0.52)^2 + (0.31)^2 + 2 \times 0.52 \times 0.31} \\ & = \sqrt{(0.52 + 0.31)^2} = (0.52 + 0.31) \\ & = 0.83 \end{aligned}$$

Sol.145.(a)

$$\begin{aligned} & \frac{49}{63} \div \frac{21}{9} - \frac{85}{510} + \frac{88}{132} \div \frac{66}{11} \\ & \frac{1}{7} \div \frac{33}{21} + \frac{26}{143} - \frac{1}{209} \div \frac{1}{19} \\ & = \frac{49}{63} \times \frac{9}{21} - \frac{85}{510} + \frac{88}{132} \times \frac{11}{66} \\ & = \frac{1}{7} \times \frac{21}{33} + \frac{26}{143} - \frac{1}{209} \times 19 \\ & = \frac{\frac{1}{3} - \frac{1}{6} + \frac{1}{9}}{\frac{1}{11} + \frac{2}{11} - \frac{1}{11}} \\ & = \frac{6-3+2}{18} \times \frac{11}{2} = \frac{55}{36} \end{aligned}$$

Sol.146.(d)

$$\begin{aligned} & \frac{\frac{5}{2} \div \frac{2}{5} \times \frac{1}{5} - \frac{1}{5} \div \frac{3}{5} \times \frac{5}{2}}{\frac{2}{5} \div \frac{1}{5} - \frac{1}{2} \times \frac{5}{2} \div \frac{25}{4} + \frac{1}{2} \div \frac{2}{5} \times \frac{1}{5}} \\ & = \frac{\frac{25}{4} \times \frac{1}{5} - \frac{1}{3} \times \frac{5}{2}}{2 - \frac{1}{2} \times \frac{2}{5} + \frac{5}{4} \times \frac{1}{5}} \\ & = \frac{\frac{5}{4} - \frac{5}{6}}{2 - \frac{1}{5} + \frac{1}{4}} = \frac{5}{12} \times \frac{20}{41} = \frac{25}{123} \end{aligned}$$

Sol.147.(a) $A = \frac{(36 \div 18 \times 5)}{(36 \div (3 \times 4))}$ and

$B = \frac{(5 \div 15 \times 5 + 6 \times 2 - 5)}{(15 \div 5 \times 25 - 2 \times 3 + 6)}$

So, $(A + B) = \frac{(36 \div 18 \times 5)}{(36 \div (3 \times 4))} + \frac{(5 \div 15 \times 5 + 6 \times 2 - 5)}{(15 \div 5 \times 25 - 2 \times 3 + 6)}$

$$= \frac{10}{3} + \frac{\frac{26}{3}}{75} = \frac{75 \times 10 + 26}{225} = \frac{776}{225}$$

Sol.148.(b) We can write the Fraction

$\frac{5}{11}, \frac{5}{7}, \frac{3}{8}, \frac{6}{13}$ as $\frac{30}{66}, \frac{30}{42}, \frac{30}{80}, \frac{30}{65}$ respectively

So, the smallest and greatest fraction respectively are $\frac{30}{80}(\frac{3}{8})$ and $\frac{30}{42}(\frac{5}{7})$

Therefore, difference of smallest and largest fractions = $\frac{5}{7} - \frac{3}{8} = \frac{19}{56}$

Sol.149.(a) from the formula,

$$\begin{aligned} & a^3 + b^3 = (a+b)(a^2 + b^2 - ab) \\ & \frac{(.91 + 0.09)(.91)^2 + (.09)^2 - .0819}{[(.91)^2 - .0819 + (.09)^2]} = 1 \end{aligned}$$

Sol.150.(a)

$$\sqrt{a - \sqrt{a - \sqrt{a - \sqrt{a - \dots \infty}}}}$$

$$= \frac{\sqrt{4a+1}-1}{2}$$

$$\sqrt{20 - \sqrt{20 - \sqrt{20 - \sqrt{20 - \dots \infty}}}} \\ = \frac{\sqrt{4 \times 20 + 1} - 1}{2} = \frac{\sqrt{81} - 1}{2} = \frac{9-1}{2} = 4$$

Short Trick:-

Co - prime factors of 20 = (4, 5)

In case of (-ve) sign, smaller number is the right answer.

Sol.151.(b)

$$\frac{\sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + \sqrt{225}}}}}}{\sqrt{16 + 19.25 \times 4^2}} \\ = \frac{\sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + 15}}}}}{\sqrt{16 + 308}} \\ = \frac{\sqrt{10 + \sqrt{25 + \sqrt{108 + 13}}}}{\sqrt{324}} \\ = \frac{\sqrt{10 + \sqrt{25 + 11}}}{18} = \frac{\sqrt{10 + 6}}{18} = \frac{2}{9}$$

Sol.152.(a) $x^9 \times x^5 \times x^{-4} \times x^0 \times x^{-6}$
 $= x^9 \times x^5 \times \frac{1}{x^4} \times 1 \times \frac{1}{x^6} \dots\dots\dots (x^0 = 1)$
 $= \frac{x^{14}}{x^{10}} = x^4 \dots\dots\dots (\text{as } \frac{p^m}{p^n} = p^{m-n})$

Sol.153.(c) Sum of 'n' natural numbers

$$= \frac{n}{2}(a + l) \dots \text{where } a = \text{first term,}$$

l = last term

$$100^2 - 99^2 + 98^2 - 97^2 + \dots + 12^2 - 11^2 \\ (100 + 99)(100 - 99) + (98 + 97)(98 - 97) + \dots + (12 + 11)(12 - 11) \\ = 100 + 99 + 98 + 97 \dots + 12 + 11 \\ \text{Sum} = \frac{90}{2}(100 + 11) = 4995$$

Sol.154.(a) $\sqrt{36x^2 - 108x + 81}$
 $= \sqrt{(6x - 9)^2} = 6x - 9$

Sol.155.(c) $\frac{\sqrt[3]{-2744} \times \sqrt[3]{-216}}{\sqrt[3]{\frac{64}{729}}}$
 $= \frac{(-14) \times (-6)}{\frac{4}{9}} = \frac{14 \times 6 \times 9}{4} = 189$

Sol.156.(b) $\{(5^6)^5 \times (5^5)^8\} \div 25 = (625)^a$
 $5^{30+40} \div 5^2 = (625)^a$
 $5^{70} \div 5^2 = (625)^a \Rightarrow 5^{68} = (625)^a$
 $(5^4)^{17} = (625)^a \Rightarrow 625^{17} = (625)^a \Rightarrow a = 17$

Sol.157.(b)

$$99 \frac{11}{99} + 99 \frac{13}{99} + 99 \frac{15}{99} + \dots + 99 \frac{67}{99} \\ = (99 + \frac{11}{99}) + (99 + \frac{13}{99}) + (99 + \frac{15}{99})$$

$$\dots\dots\dots (99 + \frac{67}{99})$$

$$= 99 \times 29 + \frac{1}{99} (11 + 13 + 15 + \dots 67) \\ = 2871 + \frac{1}{99} (29 \times \frac{67+11}{2}) \\ = 2871 + \frac{1}{99} (29 \times 39) \\ = 2871 + \frac{377}{33} = \frac{94743 + 377}{33} = \frac{95120}{33}$$

Sol.158.(c) Let $x = y = z = 2$

$$\frac{(x+y+z)(xy+yz+zx) - xyz}{(x+y)(y+z)(z+x)} \\ = \frac{(2+2+2)(2 \times 2 + 2 \times 2 + 2 \times 2) - 2 \times 2 \times 2}{(2+2)(2+2)(2+2)} \\ = \frac{(6)(4+4+4) - 8}{(4)(4)(4)} = \frac{(6)(12) - 8}{64} \\ = \frac{72 - 8}{64} = \frac{64}{64} = 1$$

Sol.159.(b)

$$\frac{\sqrt{29.16}}{\sqrt{1.1664}} + \frac{\sqrt{0.2916}}{\sqrt{116.64}} + \frac{\sqrt{0.0036}}{\sqrt{0.36}} \\ = \frac{5.4}{1.08} + \frac{0.54}{10.8} + \frac{0.06}{0.6} \\ = 5 + \frac{1}{20} + \frac{1}{10} = 5 + \frac{3}{20} = \frac{103}{20}$$

Sol.160.(c)

$$A = \frac{\sqrt{0.0004} \times \sqrt[3]{0.000008}}{\sqrt[4]{16000} \times \sqrt[3]{125000} \times \sqrt[4]{810}} \\ = \frac{0.02 \times 0.02}{2 \times 10^{\frac{3}{4}} \times 50 \times 3 \times 10^{\frac{1}{4}}} \\ = \frac{0.0004}{3 \times 10^{2+\frac{3}{4}+\frac{1}{4}}} = \frac{0.0004}{3 \times 10^3} = \frac{4}{3} \times 10^{-7} \\ B = \frac{\sqrt[3]{0.729} \times \sqrt[4]{0.0016}}{\sqrt{0.16}} = \frac{0.9 \times 0.2}{0.4} \\ = \frac{18}{40} = \frac{9}{20}$$

Now,

$$A \times B = \frac{4}{3} \times 10^{-7} \times \frac{9}{20} = \frac{3}{5} \times 10^{-7}$$

Sol.161.(a)

$$A = 0.31\overline{2} = \frac{312-3}{990} = \frac{309}{990} = \frac{103}{330} \\ B = 0.41\overline{5} = \frac{415-4}{990} = \frac{411}{990} = \frac{137}{330} \\ C = 0.30\overline{9} = \frac{309-30}{900} = \frac{279}{900} = \frac{31}{100} \\ \text{So, } A + B + C = \frac{103}{330} + \frac{137}{330} + \frac{31}{100} \\ = \frac{240}{330} + \frac{31}{100} = \frac{80}{110} + \frac{31}{100} = \frac{1141}{1100}$$

Sol.162.(a)

$$\sqrt{\frac{\sqrt{5}-1}{\sqrt{5}+1}} = \sqrt{\frac{(\sqrt{5}-1)(\sqrt{5}-1)}{(\sqrt{5}+1)(\sqrt{5}-1)}} \\ = \sqrt{\frac{(\sqrt{5}-1)^2}{4}} = \frac{(\sqrt{5}-1)}{2} = \frac{2.24-1}{2} \\ = \frac{1.24}{2} = 0.62$$

Sol.163.(d)

If $a + b + c = 0$ then $a^3 + b^3 + c^3 = 3abc$

$$\Rightarrow 13^3 + (-8)^3 + (-5)^3 = 13 \times -8 \times -5 = 0$$

$$\Rightarrow 13^3 + (-8)^3 + (-5)^3$$

$$= 3 \times 13 \times (-8) \times (-5) = 1560$$

Sol.164.(c)

$$(0.064)^{123} \div 0.16^{47} \times 0.4^{34} \times 0.4^{29} = 0.4^x \\ \Rightarrow 0.4^x = 0.4^{3 \times 123} \div 0.4^{47 \times 2} \times 0.4^{34} \times 0.4^{29} \\ \Rightarrow 0.4^x = 0.4^{369-94} \times 0.4^{34} \times 0.4^{29} \\ \Rightarrow 0.4^x = 0.4^{275} \times 0.4^{34} \times 0.4^{29} \\ \Rightarrow 0.4^x = 0.4^{275+34+29} \Rightarrow 0.4^x = 0.4^{338}$$

We know that, If bases are equal then powers will be equal.

Therefore, $x = 338$

Sol.165.(d) $P = 7 \times 4 - 3 = 25$

$$Q = 8 \div 4 + 2 = 4, R = 14 \div 21 \times 9 = 6$$

$$P \times Q + R = 25 \times 4 + 6 = 106$$

Sol.166.(c) $(x-y)(x+y) = x^2 - y^2$

$$\Rightarrow (36p^2 + 49q^2) \{(6p)^2 - (7q)^2\} \\ \Rightarrow (36p^2 + 49q^2)(36p^2 - 49q^2) \\ \Rightarrow (36p^2)^2 - (49q^2)^2 = 1296p^4 - 2401q^4$$

Sol.167.(d)

$$3p - [3p - \frac{p}{p+q} + \{3p - (p - \frac{q}{q+p})\}] \\ = 3p - [3p - \frac{p}{p+q} - \{3p - (p - q + p)\}] \\ = 3p - [3p - \frac{p}{p+q} - (3p - 2p + q)] \\ = 3p - [3p - \frac{p}{p+q} - p - q] \\ = 3p - [3p - p - q - p - q] \\ = 3p - p + 2q = 2p + 2q$$

Sol.168.(a) $(\frac{9^3 + 9^4 + 9^5 + 9^6 + 9^7}{9^1 + 9^2 + 9^3 + 9^4 + 9^5})^{\frac{1}{2}}$

$$= [\frac{9^3(1+9+9^2+9^3+9^4)}{9(1+9+9^2+9^3+9^4)}]^{\frac{1}{2}} \\ = (9^2)^{\frac{1}{2}} = 9$$

Sol.169.(b)

$$3\frac{6}{7} - \{5\frac{1}{7} - (2\frac{3}{7} - 3\frac{5}{7} - 4)\} \\ = \frac{27}{7} - \{\frac{36}{7} - (\frac{17}{7} - \frac{26}{7} - 4)\} \\ = \frac{27}{7} - \{\frac{36}{7} - (\frac{17}{7} - \frac{26}{7} + 4)\} \\ = \frac{27}{7} - \{\frac{36}{7} - (-\frac{9}{7} + 4)\} \\ = \frac{27}{7} - \{\frac{36}{7} - (\frac{19}{7})\} = \frac{27}{7} - \{\frac{17}{7}\} = \frac{10}{7}$$

Sol.170.(c) $\sqrt{a^{-1}b} \times \sqrt{b^{-1}c} \times \sqrt{c^{-1}a}$

$$= \sqrt{\frac{b}{a}} \times \sqrt{\frac{c}{b}} \times \sqrt{\frac{a}{c}} \\ = \sqrt{\frac{b}{a} \times \frac{c}{b} \times \frac{a}{c}} = 1$$

Sol.171.(d) Formula used :-

$$(x^3 + y^3 + z^3) = 3xyz \text{ (when } x + y + z = 0)$$

$$\frac{(a-b)^3 + (b-c)^3 + (c-a)^3}{(a^2-b^2)^3 + (b^2-c^2)^3 + (c^2-a^2)^3}$$

$$= \frac{3(a-b)(b-c)(c-a)}{3(a^2-b^2)(b^2-c^2)(c^2-a^2)}$$

$$= \frac{3(a-b)(b-c)(c-a)}{3(a+b)(a-b)(b+c)(b-c)(c+a)(c-a)}$$

$$= \frac{1}{(a+b)(b+c)(c+a)}$$

Sol.172.(a) $\sqrt{2+\sqrt{3}} + \sqrt{2-\sqrt{3}}$

$$= \sqrt{\frac{2(2+\sqrt{3})}{2}} + \sqrt{\frac{2(2-\sqrt{3})}{2}}$$

$$= \sqrt{\frac{4+2\sqrt{3}}{2}} + \sqrt{\frac{4-2\sqrt{3}}{2}}$$

$$= \sqrt{\frac{(\sqrt{3}+1)^2}{2}} + \sqrt{\frac{(\sqrt{3}-1)^2}{2}}$$

$$= \frac{\sqrt{3}+1}{\sqrt{2}} + \frac{\sqrt{3}-1}{\sqrt{2}} = \frac{2\sqrt{3}}{\sqrt{2}} = \sqrt{6}$$

Sol.173.(b) Formula used :-

$$a^3 + b^3 + c^3 = 3abc \text{ when } a+b+c=0$$

$$\frac{(r-s)^3 + (s-t)^3 + (t-r)^3}{6(r-s)(s-t)(t-r)}$$

$$= \frac{3(r-s)(s-t)(t-r)}{6(r-s)(s-t)(t-r)} = \frac{3}{6} = \frac{1}{2}$$

Sol.174.(d) $(a^{-1} + b^{-1}) \div (a^{-3} + b^{-3})$

$$= \left(\frac{1}{a} + \frac{1}{b}\right) \div \left(\frac{1}{a^3} + \frac{1}{b^3}\right)$$

$$= \left(\frac{a+b}{ab}\right) \div \left(\frac{a^3+b^3}{a^3b^3}\right)$$

$$= \left(\frac{a+b}{ab}\right) \times \left(\frac{a^3b^3}{a^3+b^3}\right) = \frac{a^2b^2}{a^2-ab+b^2}$$

Sol.175.(b) Put $a=b=1$ and $c=0$.

$$\frac{(a^2-4b^2)^3 + 64(b^2-4c^2)^3 + (16c^2-a^2)^3}{(a-2b)^3 + (2b-4c)^3 + (4c-a)^3}$$

$$\frac{(1^2-4 \times 1^2)^3 + 64(1^2-4 \times 0^2)^3 + (16 \times 0^2-1^2)^3}{(1-2)^3 + (2-4 \times 0)^3 + (4 \times 0-1)^3}$$

$$= \frac{36}{6} = 6$$

On checking the option we get only option (b) satisfies this.

Sol.176.(c) Given that

$$5\sqrt{3} + \sqrt{75} = 17.32$$

$$5\sqrt{3} + 5\sqrt{3} = 10\sqrt{3} = 17.32$$

Now according to question

$$14\sqrt{3} + \sqrt{108} = 14\sqrt{3} + 6\sqrt{3} = 20\sqrt{3}$$

$$2 \times 17.32 = 34.64$$

Sol.177.(a) $847 = 7^1 \times 11^2$, $385 = 5 \times 7 \times 11$

$$675 = 3^3 \times 5^2$$
, $3025 = 5^2 \times 11^2$

$$847 \times 385 \times 675 \times 3025 = 3^a \times 5^b \times 7^c \times 11^d$$

$$3^3 \times 5^5 \times 7^2 \times 11^5 = 3^a \times 5^b \times 7^c \times 11^d$$

On comparing, we get

$$a=3, b=5, c=2, d=5$$

$$ab - cd = 3 \times 5 - 2 \times 5 = 5$$

Sol.178.(a)

$$\frac{1}{4-\sqrt{15}} = \frac{(4+\sqrt{15})}{(4-\sqrt{15})(4+\sqrt{15})} = 4+\sqrt{15}$$

Similarly, we get conjugate of all values,

$$4+\sqrt{15}-\sqrt{15}-\sqrt{14}+\sqrt{14}+\sqrt{13}-\sqrt{13}-\sqrt{12}$$

$$+\sqrt{12}+\sqrt{11}-\sqrt{11}-\sqrt{10}+\sqrt{10}+3-3-\sqrt{8}$$

$$\Rightarrow 4-\sqrt{8} = 4-2\sqrt{2}$$

Sol.179.(c) $\frac{\sqrt{38-5\sqrt{3}}}{\sqrt{26+7\sqrt{3}}}$

on multiplying numerator and denominator by $\sqrt{2}$ we get,

$$\frac{\sqrt{76-10\sqrt{3}}}{\sqrt{52+14\sqrt{3}}} = \frac{\sqrt{(5\sqrt{3}-1)^2}}{\sqrt{(7+\sqrt{3})^2}} = \frac{5\sqrt{3}-1}{7+\sqrt{3}}$$

$$\frac{(5\sqrt{3}-1)(7-\sqrt{3})}{(7+\sqrt{3})(7-\sqrt{3})} = \frac{-11+18\sqrt{3}}{23}$$

On comparing $\frac{-11+18\sqrt{3}}{23}$

with $\frac{a+b\sqrt{3}}{23}$ we get,

$$\Rightarrow a = -11 \text{ and } b = 18$$

$$(b-a) = 18 - (-11) = 29$$

Sol.180.(a) $\{4 - \frac{2}{1+\frac{2}{1-\frac{1}{2+\frac{3}{4}}}}\} \div \frac{5}{12}$

$$\text{of } \frac{72}{145} - (4 + 3 \div 0.5 - 1)$$

$$= \{4 - \frac{2}{1+\frac{2}{1-\frac{1}{11}}}\} \div \frac{17}{12} \times \frac{72}{45} - (4 +$$

$$6-1) = \{4 - \frac{2}{1+\frac{22}{7}}\} \div \frac{102}{145} - (9)$$

$$= \{4 - \frac{14}{29}\} \div \frac{102}{145} - (9)$$

$$= \{\frac{102}{29}\} \div \frac{102}{145} - (9)$$

$$= \{\frac{102}{29}\} \times \frac{145}{102} - (9) = 5-9 = -4$$

Sol.181.(d) $4\frac{2}{5} \div \{(1\frac{1}{2} - 3\frac{1}{5}) \div 3\frac{2}{5}$

$$+ (2\frac{1}{5} \div 1\frac{1}{2} + 4\frac{2}{5}) + \frac{1}{2}\}$$

$$= \frac{22}{5} \div \{(\frac{3}{2} - \frac{16}{5}) \div \frac{17}{5} + (\frac{11}{5} \div \frac{3}{2}$$

$$+ \frac{22}{5}) + \frac{1}{2}\}$$

$$= \frac{22}{5} \div \{(-\frac{17}{10}) \times \frac{5}{17} + (\frac{11}{5} \times \frac{2}{3} +$$

$$\frac{22}{5}) + \frac{1}{2}\}$$

$$= \frac{22}{5} \div \{-\frac{1}{2} + (\frac{88}{15}) + \frac{1}{2}\}$$

$$= \frac{22}{5} \div \{\frac{88}{15}\} = \frac{22}{5} \times \frac{15}{88} = \frac{3}{4}$$

Sol.182.(a)

$$3\frac{5}{6} + [3\frac{2}{3} + \{\frac{15}{4} - (5\frac{4}{5} \div 14\frac{1}{2})\}]$$

$$= \frac{23}{6} + [\frac{11}{3} + \{\frac{15}{4} - (\frac{29}{5} \div \frac{29}{2})\}]$$

$$= \frac{23}{6} + [\frac{11}{3} + \{\frac{15}{4} \times \frac{2}{5}\}]$$

$$= \frac{23}{6} + [\frac{11}{3} + \frac{3}{2}] = \frac{23}{6} + \frac{31}{6} = 9$$

Sol.183.(c)

$$(\frac{3}{4} - \frac{1}{4} \div \frac{1}{4} \text{ of } \frac{2}{5}) \div (\frac{3}{4} \div \frac{2}{3} \text{ of } \frac{3}{5})$$

$$= (\frac{3}{4} - \frac{1}{4} \div \frac{1}{10}) \div (\frac{3}{4} \div \frac{2}{5})$$

$$= -\frac{7}{4} \div (-\frac{15}{8}) = -\frac{14}{15}$$

Sol.184.(b) $(2x-3y)^3 - 18xy(2x-3y)$

$$= 8x^3 - 27y^3 - 36x^2y + 54xy^2 - 36x^2y + 54xy^2$$

$$= 8x^3 - 27y^3 - 72x^2y + 108xy^2$$

Sol.185.(a)

$$364 - [142 - \{75 + (38 - \frac{5}{4} + \frac{14}{8})\}]$$

$$364 - [142 - \{75 + (38 - 3)\}] = 332$$